The power behind competitiveness

Grid-tie Transformerless Solar Inverter

M88H

Operation and Installation Manual

www.deltaww.com

Smarter. Greener. Together.

Table of Contents

Figure

Table

1 Safety

1.1 Information of the Inverter

1.1.1 Disclaimer

This manual accompanies our product for use by the end users. The technical instructions and illustrations contained in this manual are to be treated as confidential and no part may be reproduced without the prior written permission of DELTA ELECTRONICS, INC. Service engineers and end users may not divulge the information contained herein or use this manual for purpose other than those strictly connected with correct use of the product. All information and specifications are subject to change without notice.

DELTA ELECTRONICS, INC. shall have no obligation to both personal injury and property damage hereinafter with respect to any actions -- (a) the product has been installed and repaired improperly; (b) the product has been misuse without following the instructions on this user manual; (c) the product has failed due to incorrect unpacking.

1.1.2 Target Group

This user manual of the solar inverter is prepared for a person who is well-trained for installing, commissioning, using, and doing maintenance. The well-trained person must have the following basic and advanced skills:

- The fundamentals of electricity, wiring, electrical components and electrical schematic symbols.
- Knowledge of how a solar inverter works and is operated.
- Training in the installation and commissioning of electrical devices and installations.
- Training in how to deal with the dangers and risks associated with installing and using electrical devices and installations.
- Compliance with this manual and all safety information.

Please read the user manual before working on the product.

1.2 General Safety

IMPORTANT SAFETY INSTRUCTIONS : SAVE THESE INSTRUCTIONS !

- Please read these instructions carefully and save them for later use.

To prevent any personal injury and any property damage, also ensure long-term operation of the solar inverter, you must read this section carefully and review all the safety instructions at all times before using this inverter.
This user manual provides important instructions for Delta grid-tie transformerless solar inverter. The product is designed, tested, verified, and certified according to international safety requirements, regulations, and standards but precautions must be observed when installing and operating the product.

ATTENTION : NO GALVANIC ISOLATION

- There is no accessory such as a transformer along with our product and therefore the product has no galvanic isolation.
  External transformer should be installed between AC output of inverter and Grid.
  Please do not connect grounded Photovoltaic modules to the product.
  If you connect grounded Photovoltaic modules to the product, the error message **INSULATION (E34)** shows up.
- It is prohibited to connect the L1, L2, L3, and N to the ground.

1.2.1 Condition of Use

The M88H is a transformerless solar inverter with two MPP trackers which converts the variable direct current of the solar array into a utility frequency grid-compliant three-phase current and feeds it into the utility grid.
The Photovoltaic modules used must be compatible with the inverter.
Photovoltaic modules with a high parasitic capacitance to ground may only be applied if the capacitive coupling does not exceed 8μF.

The inverter must only be operated in countries for which it is approved by Delta and the grid operator.

1.2.2 Symbols

This section describes the definition of the symbols in this manual.
In order to prevent both personal injury and property damage, and to ensure long-term operation of the product, please read this section carefully and follow all the safety instructions while you use the product.

DANGER!

- This warning indicates an immediate hazard which will lead to death or serious injury may occur.

WARNING !

- This warning indicates a possible hazard which may lead to death or serious injury may occur.

CAUTION !

- This warning indicates a possible hazard which may lead to minor injury may happen.

ATTENTION

- This warning indicates a possible damage to property and the environment might happen.

INFORMATION

- Additional information is indicated by an exclamation mark enclosed by double circle. This means the following section contains important information and user should follow the instruction to prevent any hazards.

DANGER : ELECTRICAL HARZARD!!

- This warning indicates an immediate electrical hazard which will lead to death or s erious injury may occur.

CAUTION : HOT SURFACES, DO NOT TOUCH!

- This warning indicates be careful of hot surfaces when operating the product.
- Do not perform any task until the product cool down sufficiently.

- Wait for a prescribed amount of time before engaging in the indicated action.
- Patientez le délai requis avant d’entreprendre l’action indiquée

- Equipment grounding conductor (PE)
- (PE) Équipement conducteur de terre

2 Introduction

The M88H is designed to enable the highest levels of efficiency and provide longest operating life of photovoltaic inverter by state-of-the-art high-frequency and low EMI technology. It is suitable for outdoor use.

ATTENTION : NO GALVANIC ISOLATION

- There is no accessory such as a transformer along with our product and therefore the product has no galvanic isolation.
  External transformer should be installed between AC output of inverter and Grid.
  Please do not connect grounded Photovoltaic modules to the product.
  If you connect grounded Photovoltaic modules to the product, the error message **INSULATION (E34)** shows up.
- It is prohibited to connect the L1, L2, L3, and N to the ground.

2.1 Valid Model

The user manual is valid for the following device types :

• M88H_121
• M88H_122

This user manual must be followed during installation, operation, and maintenance.

The M88 Series have 2 models as shown in ***Figure 2-2.*** Delta reserves the right to make modifications to the content and technical data in this user manual without prior notice.

2.2 Product Overview

The components of M88H is shown as ***Figure 2-1***.

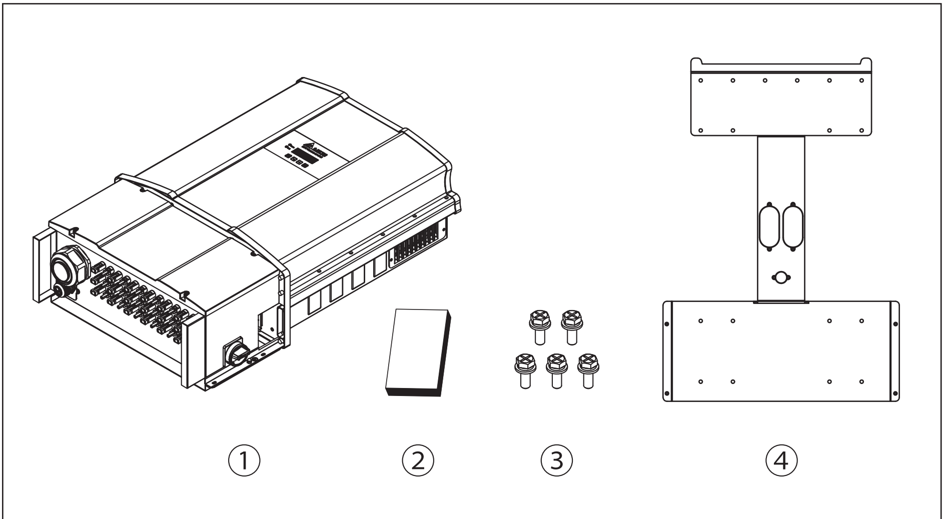


Figure 2-1 : Components of M88H

M88H			
	Object	Qty	Description
1	Delta Solar Inverter	1 pc	Solar inverter
2	User Manual	1 pc	Important instructions for solar inverter Safety instructions should be followed during installation and maintenance
3	Screw	5 pcs	Inverter with wall mount screws
4	Mounting Bracket	1 pc	Wall mounting bracket (indude packing for wiring box)

Table 2-1 : Packing list of M88H

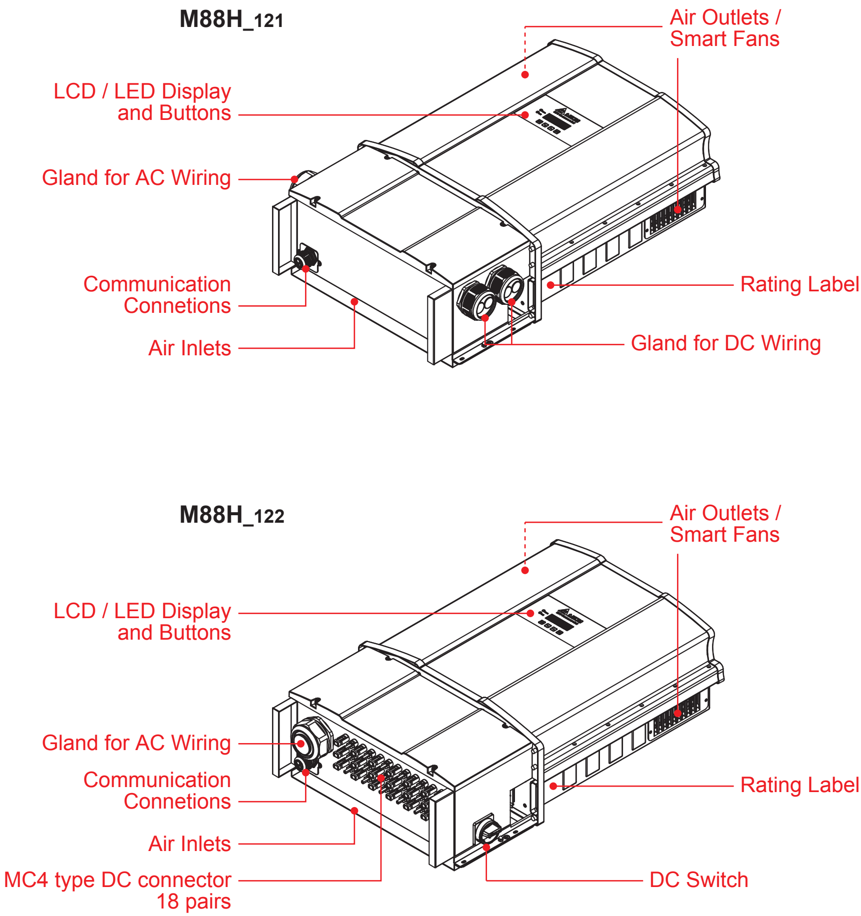
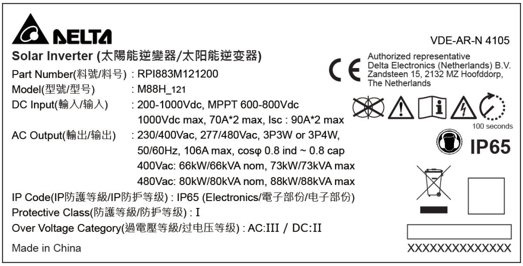


Figure 2-2 : Overview of M88 series

The following ***Figure 2-3*** shows the rating labels of M88H along with these labels explains the definition of the specific mark.
Figure 2-4 illustrates the layout of wiring box of M88H and the table (***Table 2-3***) along with this layout describes the detail of each area.
This compartment includes inverter inputs (DC), outputs (AC), surge protection device (SPD), fuse holder, DC switch and the communication connection such as RS-485.



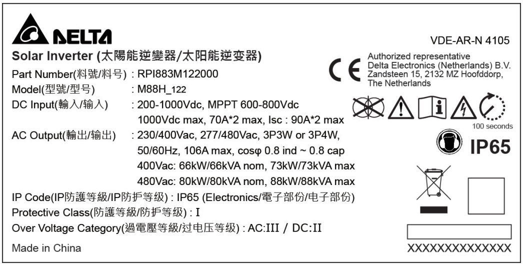


Figure 2-3 : Rating labels of M88H

Symbol	Definition
 100 seconds	**Danger to life through electric shock** Potentially fatal voltage is applied to the inverter during operation. This voltage persists even 100 seconds after disconnection of the power supply. Never open the inverter. The inverter contains no components that must be maintained or repaired by the operatoror installer. Opening the housing will void the warranty.
	Before working with the inverter, you must read the supplied manual and follow the instructions contained therein.
	This inverter is not separated from the grid with a transformer.
	The housing of the inverter must be grounded if this is required by local regulations.
	Please be aware of noise protection.
	**WEEE marking** The inverter must not be disposed of as standard household waste, but in accordance with the applicable electronic waste disposal regulations of your country or region.

Table 2-2 : Rating label explanation of M88H

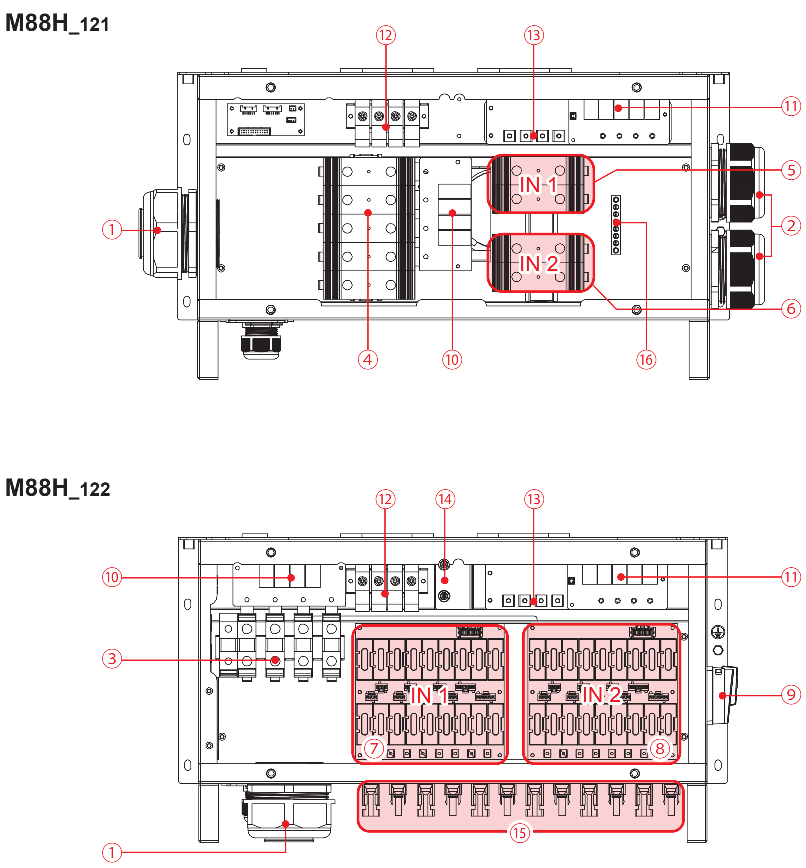


***Figure 2-4** : Wiring box layout of M88H*

NO.	Component	NO.	Component	NO.	Component
1	Cable gland for AC	7	Fuse holder type DC IN1	13	Internal DC terminal
2	Cable gland for DC	8	Fuse holder type DC IN2	14	Wiring box fan
3	95mm^2 AC terminal	9	DC switch	15	MC4 connector
4	120mm^2 AC terminal	10	Type II AC SPD	16	Grounding bar
5	120mm^2 DC terminal for IN 1	11	Type II DC SPD		
6	120mm^2 DC terminal for IN 2	12	Internal AC terminal		

Table 2-3 : Wiring box layout description of M88H

3 Installation

CAUTION !

- The unit should not be installed in direct sunlight.

WARNING !

- Do not install the unit near or on flammable surfaces.
- Please mount the unit tightly on a solid / smooth surface.

The chapter contains instructions for (1) Mechanical installation; (2) Electrical Installation; (3) Communication setup.

3.1 Mechanical Installation

This unit is designed to be wall-mounted. Please ensure that the installation is perpendicular to the floor and the AC and DC terminal are at the bottom position. Please follow the instruction as shown from ***Figure 3-1*** through ***3-3***. First, fix the wall mounting bracket on a solid support surface. Second, please mount the inverter on the bracket securely. Note that ***Figure 3-1*** through ***3-7*** should be followed.

To mount the inverter on the wall, please follow the procedure :

1. Screw the wall mounting bracket on the wall with at least 8 M8 Phillips head screws. Please refer to ***Figure 3-1*** and ***3-2*** for correct installation.
2. Hang the inverter on the wall mounting bracket.
3. There are two ways for fixing the wall mounting bracket as shown in ***Figure 3-2***.
4. ***Figure 3-3*** through ***3-7*** describes correct mounting installation.
5. ***Figure 3-7*** shows the installation detail for fixing the wiring box.

CAUTION !

- Please fix at least 8 M8 Phillips head screws for the wall mounting bracket.
- The bracket shipped with the unit is specially designed and should be the only mounting device for the mechanical installation.

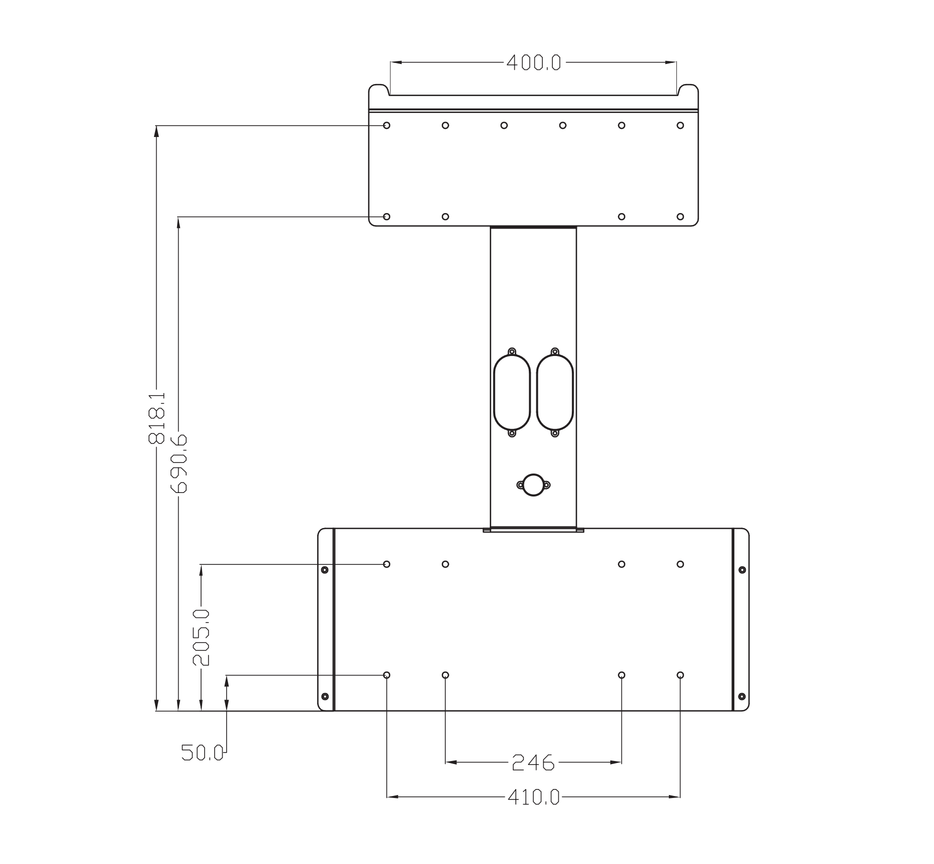


Figure 3-1** : **Mounting bracket dimensions

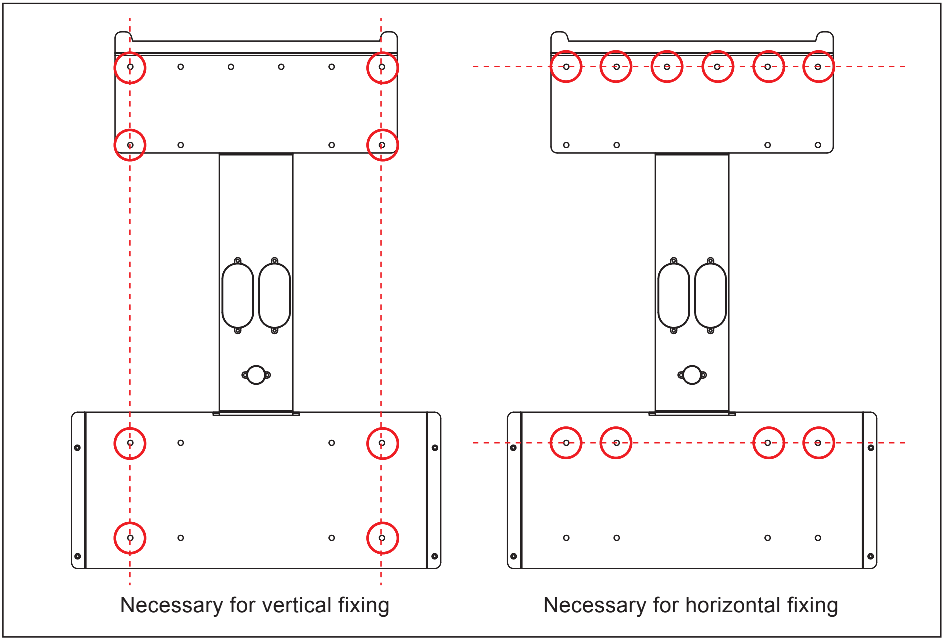


***Figure 3-2** : Required position for at least 8 screws*

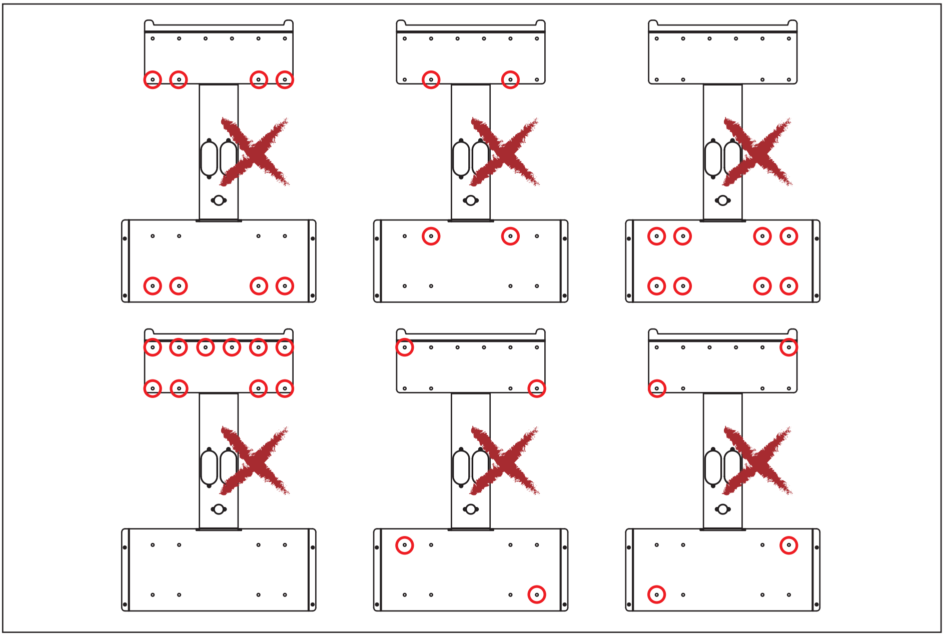


***Figure 3-3** : Prohibited position for screws*

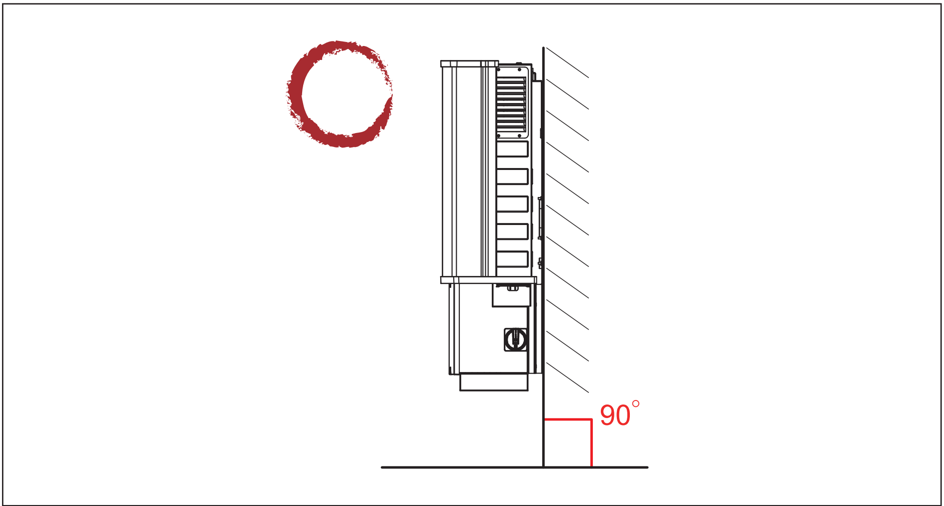


***Figure 3-4** : Permitted mounting positions*

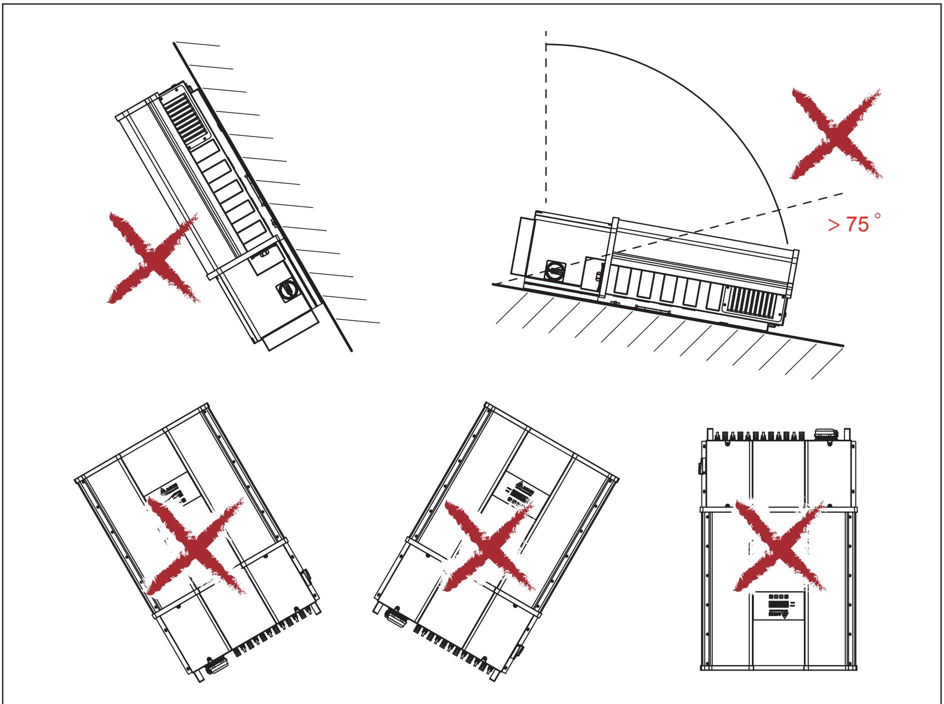


***Figure 3-5** : Prohibited mounting positions*

O : Permitted / X : Prohibited

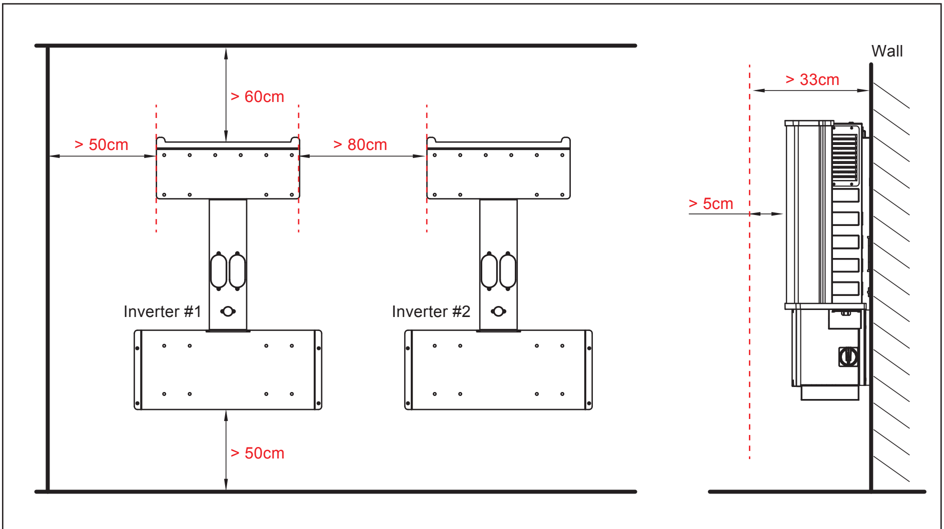


Figure 3-6** : **Required mounting clearances

CAUTION !

- Please follow the instructions above such as permitted positions and permitted mounting clearances for the correct installation.

After installing the unit, fix the wiring box with 4 screws.
The torque of the screw: 45 kgf.cm

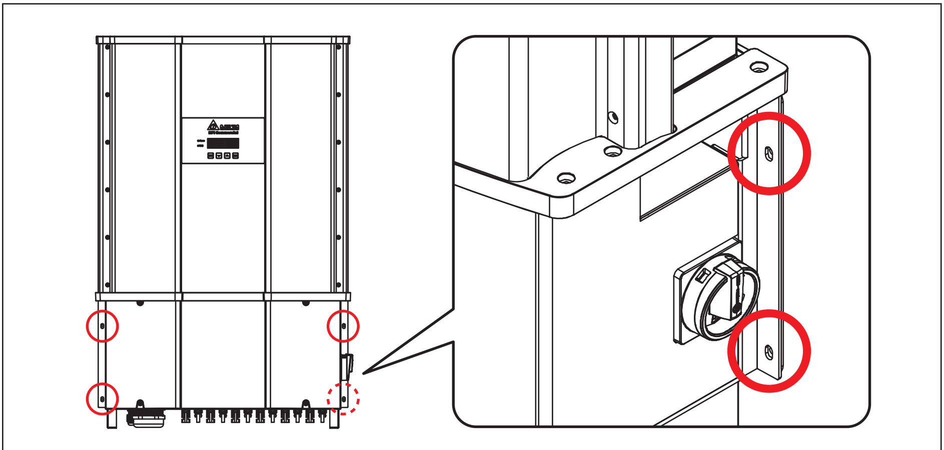


Figure 3-7** : **Specification of fixing WB for wall-mounting installation

3.2 Electrical Installation for AC Cabling

DANGER : ELECTRICAL HARZARD!!

- Any AC electric power connected to the inverter during cabling is prohibited.

WARNING !

- Code compliance is the installer's responsibility.
- Inverter warranty void if the DC input voltage exceeds 1000 Vdc.

CAUTION : INVERTER AND EQUIPMENT DAMAGE MAY OCCUR !

- Please make sure to choose the proper size for AC cable.
- Installation for AC terminal must meet the local electrical code.
- Failed to follow the instructions may damage AC cable.

ATTENTION

- This inverter may be damaged due to moisture or dust intrusion.
  Please do not open the lid of the inverter.

3.2.1 Required Protective Devices and AC Cabling Installation for M88H_121

It is recommended to install an upstream circuit breaker between AC side and inverter side for over current protection.

Model	Upstream circuit breaker
M88H	≧125A

Please follow the following steps for assembling the AC terminal (M88H_121) :

• It is important to choose the proper size for AC cable.
• Strip off all wires for 40 mm.
• The cross-sectional area for each internal cable is 1AWG ~ 250kcmil.

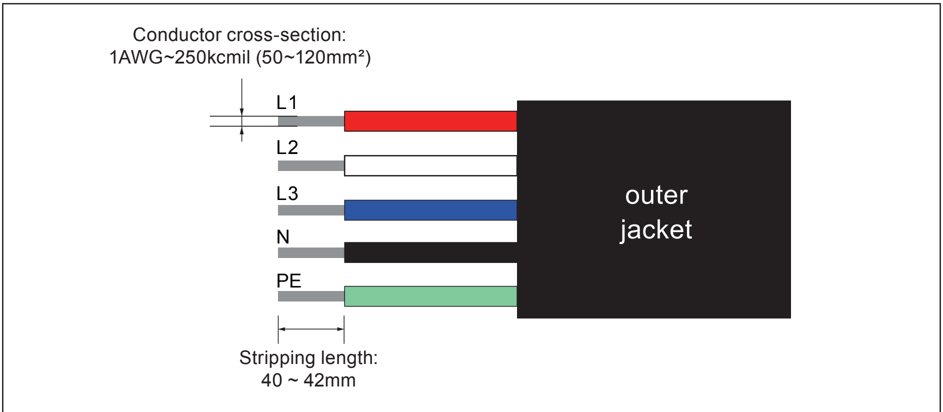


Figure 3-8 : Stripping cables for M88H_121 AC terminal

- It is important to choose the correct fixing screws for the photovoltaic fuseholder.
- Each individual screw with torque 26 N.m is required.

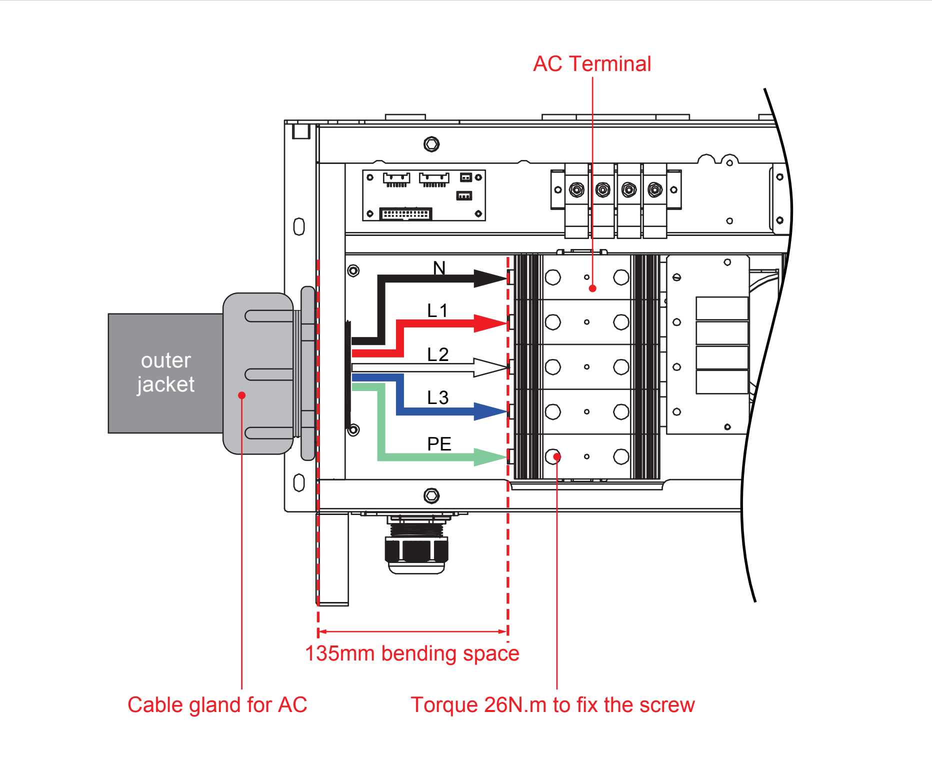


Figure 3-9 : Location for M88H_121 AC terminal

3.2.2 Required Protective Devices and AC Cabling Installation for M88H_122

It is recommended to install an upstream circuit breaker between AC side and inverter side for over current protection.

Model	Upstream circuit breaker
M88H	≧125A

Please follow the following steps for assembling the AC terminal (M88H_122) :

• It is important to choose the proper size for AC cable.
• Strip off all wires for 24 mm.
• The cross-sectional area for each internal cable is 2~3/0 AWG.

Figure 3-10 : Stripping cables for M88H_122 AC terminal

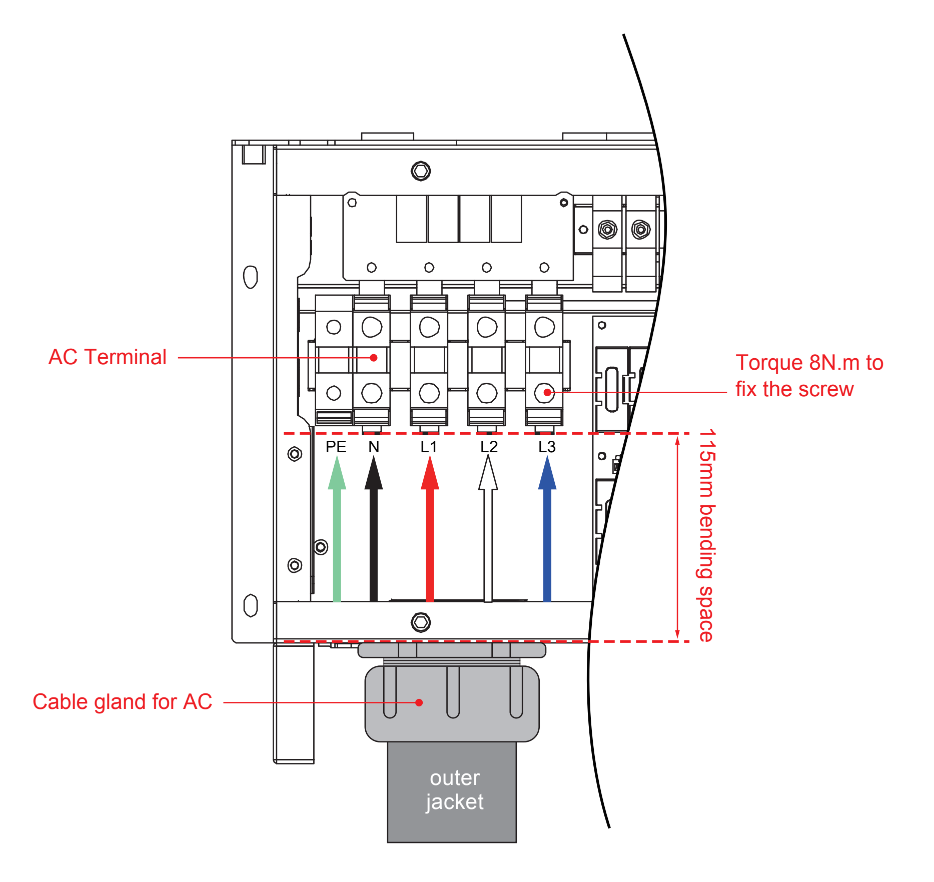


Figure 3-11 : Location for M88H_122 AC terminal

Figure 3-15 illustrates the location of cable gland for AC terminal as shown above :

- It is important to choose for the fixing screws.
- Each individual screw with torque 8N.m is required
- The bending space is 115 mm which meets the safety standard.

Please follow the following guideline for cabling if you use aluminum cables.

Guideline for aluminum conductor :
- The oxide layer must be removed from the surface of the stripped aluminum conductor.
- The stripped aluminum conductor is greased with Vaseline or contact grease with comparable properties after oxide layer removed.
- Tightened with the maximum tightening torque for the modular terminal block.
- The installation location must be kept free from humidity or aggressive atmospheres.
- It is recommend to apply to sector-shaped single-strand conductor ;
  The conductor shape must match the sector-shaped connection

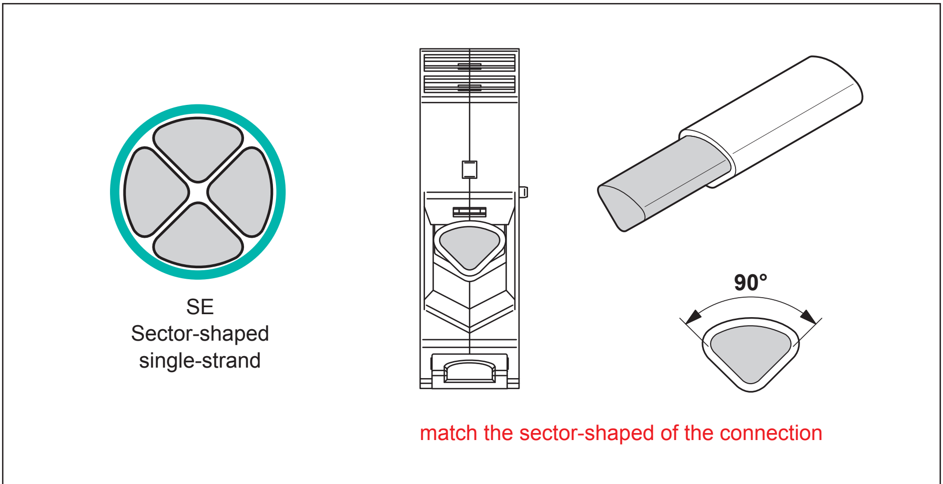


Figure 3-12 : Guideline for aluminum conductor

3.3 Electrical Installation for DC Cabling

DANGER : ELECTRICAL HARZARD!!

- PV array converts sunlight into electric power with high DC voltage and high DC current which can cause dangerous electrical shock hazard. Please use an opaque material to cover the PV array before wiring or cabling.
- Please ensure the correct polarities are connected when DC cabling is applied.

WARNING !

- The risk of electric shock and fire. Only PV modules that are listed with system voltage under 1100V are permitted for use.
- Please ensures that the DC Switch Turns "OFF" as well as the PV array is disconnected when DC cabling is applied.

ATTENTION

- The PV Array positive or negative leads must not be connected to ground.

3.3.1 DC Cabling Installation for M88H_121

Please read the following instructions for attaching DC terminals (M88H_121) :

- It is important to choose the proper size for DC cable.
- The cross-sectional area for each internal cable is 1AWG~250kcmil.

- It is important to choose the correct fixing screws.
- Each individual screw with torque 26N.m is required.
- The bending space is 135 mm which meets the safety standard.

- DC Terminals connection as seen in ***Figure 3-14.***

ATTENTION

- The screw with torque 26N.m is required for fixing.
- The required bending space is 135 mm as the requirement states.

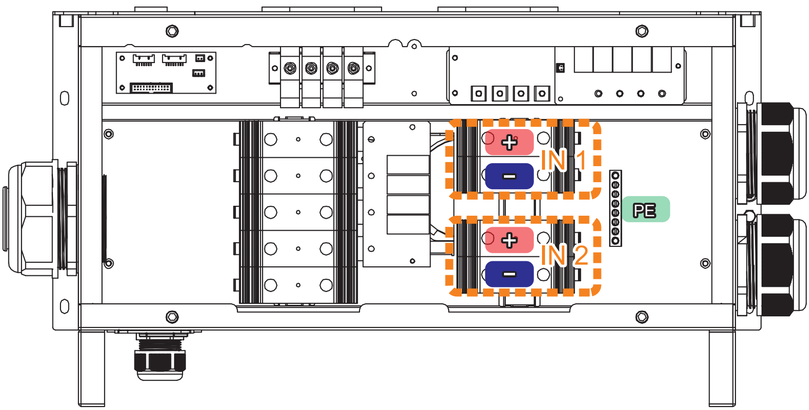


Figure 3-13 : Wiring Box layout for M88H_121

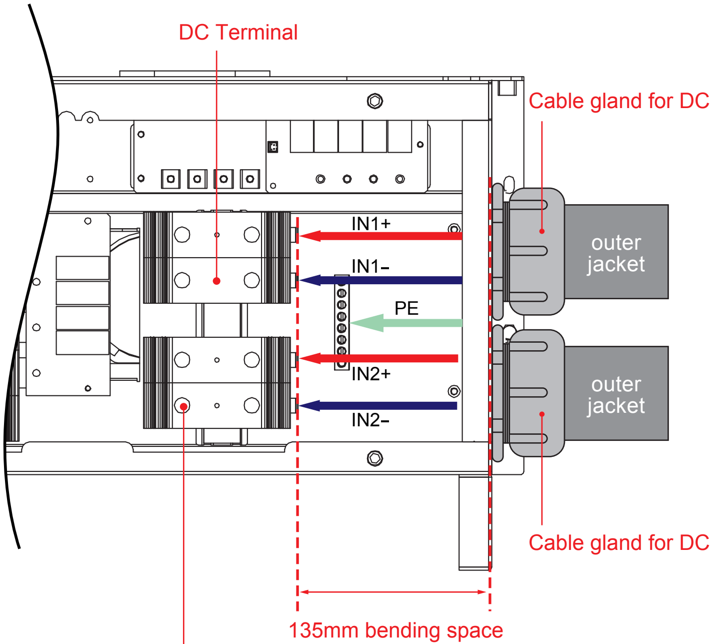


Figure 3-14 : Location for DC terminals for M88H_121

3.3.2 DC Cabling Installation for M88H_122

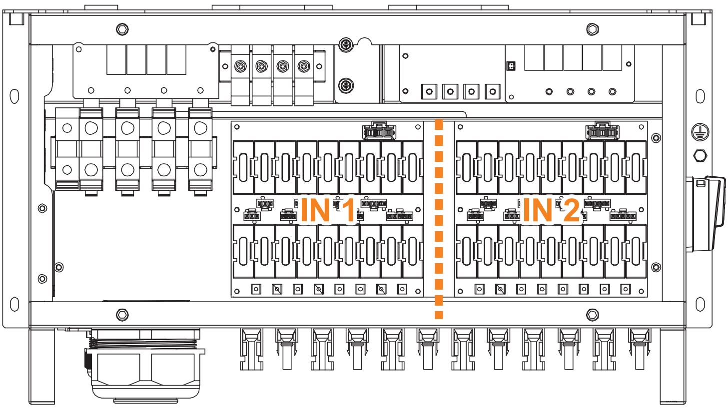


Figure 3-15 : Wiring Box layout for M88H_122

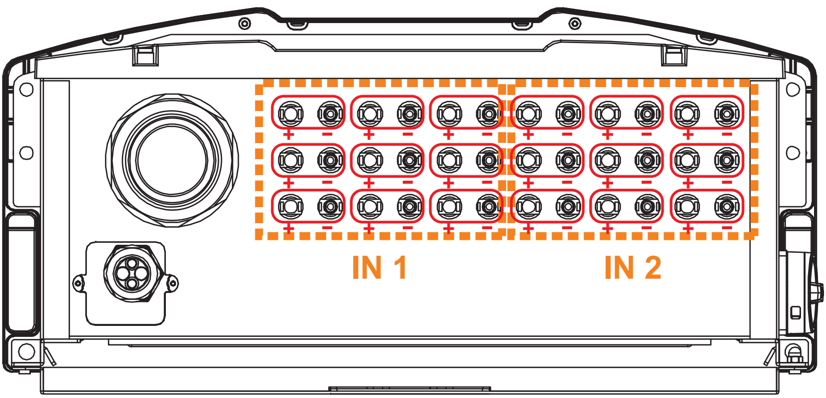


Figure 3-16 : Location for DC terminals for M88H_122

Please follow the following guideline for cabling if you use aluminum cables.

Guideline for aluminum conductor :

- The oxide layer must be removed from the surface of the stripped aluminum conductor.
- The stripped aluminum conductor is greased with Vaseline or contact grease with comparable properties after oxide layer removed.
- Tightened with the maximum tightening torque for the modular terminal block.
- The installation location must be kept free from humidity or aggressive atmospheres.
- It is recommend to apply to sector-shaped single-strand conductor ; The conductor shape must match the sector-shaped connection

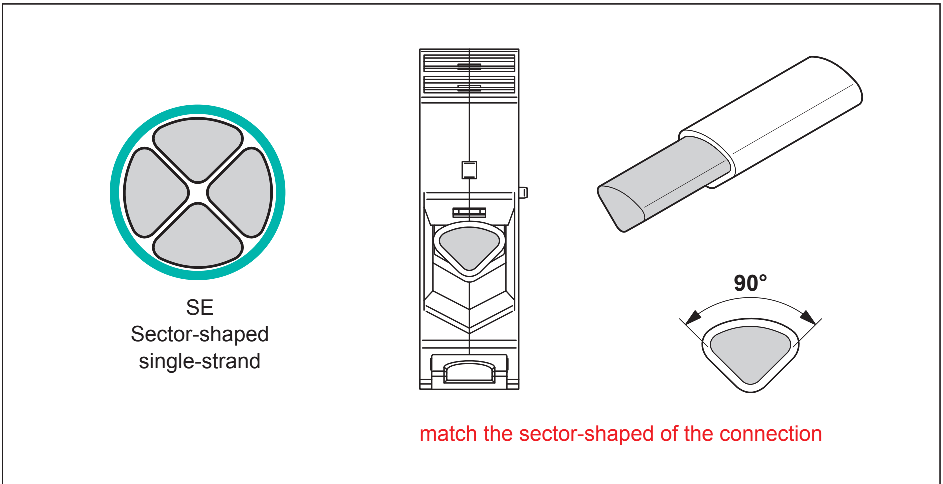


Figure 3-17 : Guideline for aluminum conductor

3.4 Communication Module Connections

The communication module of M88H provides VCC, RS-485, dry contact, EPO, and Digital Input terminals for different use.

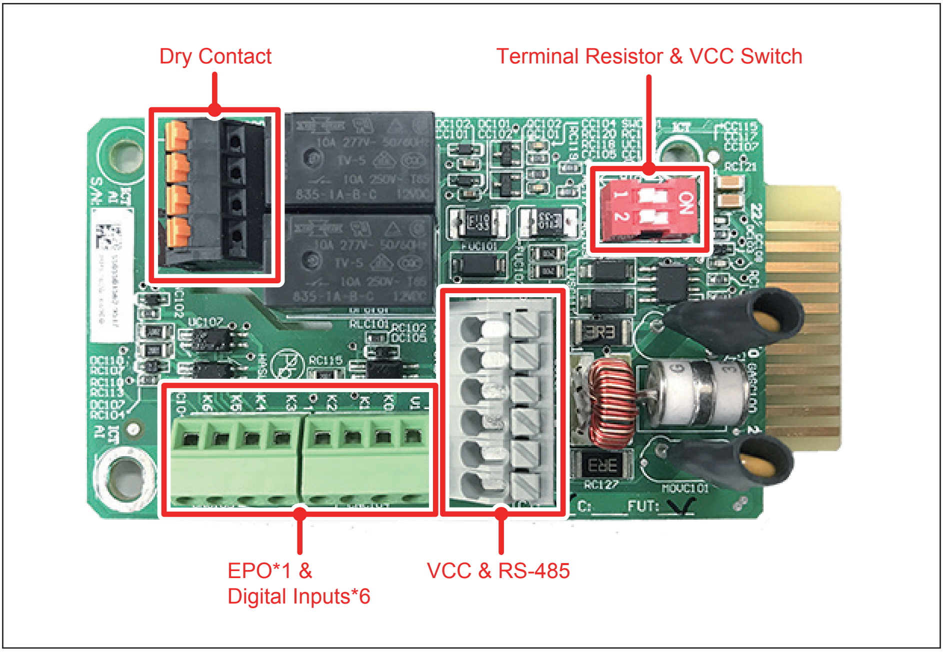


Figure 3-18 : Communication Module

3.4.1 RS-485 Connection

The pin definition of RS-485 is shown in the following table. Different RS-485 connection requires different set up for the terminal resistor.

- When single inverter is installed, the terminal resistor on its communication module should be switched ON.
- When several inverters are cascaded, only the first and the last inverter’s terminal resistors MUST be switched ON.

Pin	Function
1	VCC (+12V)
2	GND
3	DATA+
4	DATA-
5	DATA+
6	DATA-

Table 3-1 : Definition of RS-485

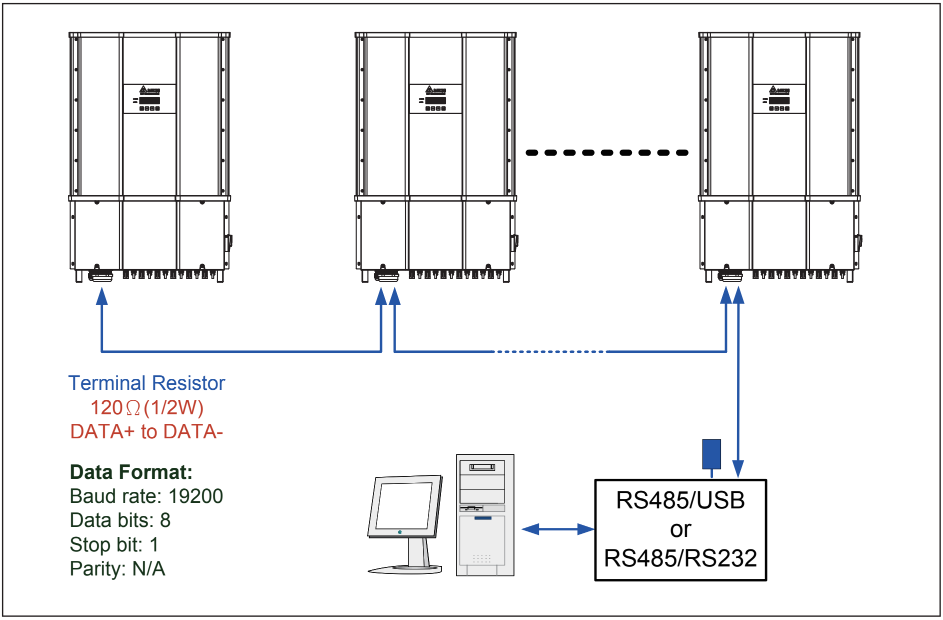


Figure 3-19 : Multiinverter connection illustration

	Switch 1	Switch 2
ON	VCC ON	Terminal Resistor ON
OFF	VCC OFF	Terminal Resistor OFF

Table 3-2 : Terminal resister setting

3.4.2 EPO Function & Digital Input

Communication Module has 1 set of emergency power off function (EPO). Users can customize EPO function in Install Settings page.

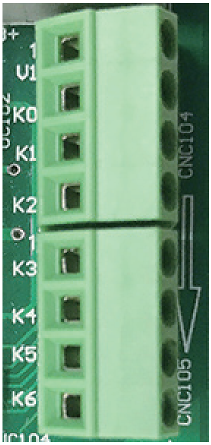


Figure 3-20 : EPO function

Short	Inverter's action
V1 & K0	Emergency power off (EPO)
V1 & K1	0% active power
V1 & K2	Maximum 30% rated power
V1 & K3	Maximum 60% rated power
V1 & K4	Maximum 100% rated power
V1 & K5	Reserved
V1 & K6	Reserved

Table 3-3 : Definition of digital input & EPO function

3.4.3 Dry Contact connection

M88H provide 2 sets of Dry Contact. The function can be customized by users, please refer to section 4.2.10 Dry Contact.

The dry contact port can withstand with 250Vac/28Vdc/9A, and suitable electric wire is 0.2-1.5mm^2.

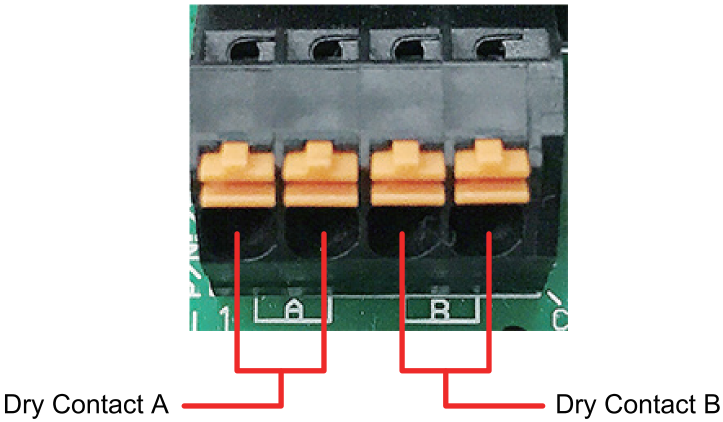


Figure 3-21 : Dry Contact connection

4 Commissioning

CAUTION : HOT SURFACES, DO NOT TOUCH!

- This warning indicates be careful of hot surfaces when operating the product.
- Do not perform any task until the product cool down sufficiently.

4.1 Display Introduction

M88 series include a 4x20 character type LCD display and 2 LED lights (located on the left-hand side of the LCD) to indicate inverter's status as shown in ***Figure 4-1***. Please refer to ***Table 4-1*** for more information about inverter's statuses and LED indicator.

The following section will introduce the functions that can be adjusted by users through the LCD panel. When you are adjusting settings, LCD panel will change the display cursor from "►" to " ⇥ " .

Power Meter	4.2.2
Energy Log	4.2.3
Event Log	4.2.4
Inverter Information	4.2.5
General Settings	4.2.6
Install Settings	4.2.7
Inverter ID	4.2.8
Insulation	4.2.9
Dry Contact	4.2.10
PID	4.2.11

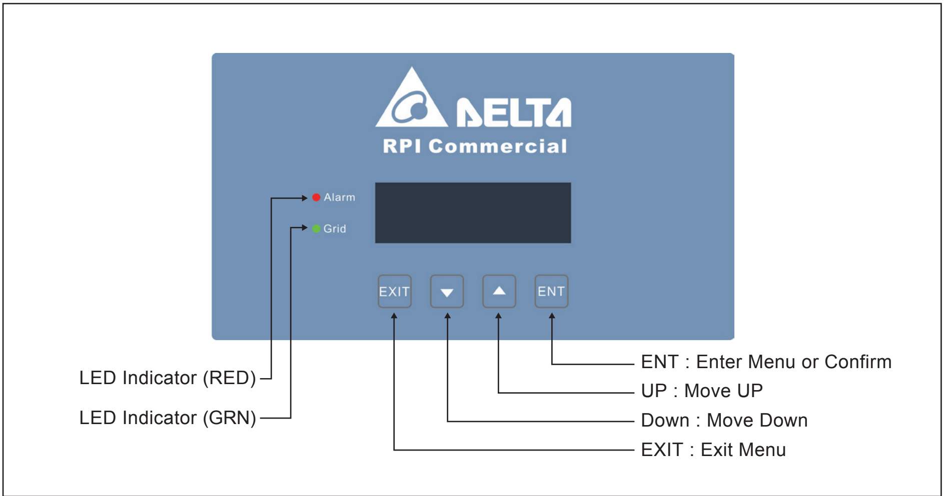


Figure 4-1 : Panel indicator

Condition	Green LED	Red LED
Standby or Countdown	FLASH *	OFF
Power ON	ON	OFF
Error or Fault	OFF	ON
Night time (No DC)	OFF	OFF
Bootloader mode	FLASH *	

* ON 1s / OFF 1s

Table 4-1 : LED indicator

4.2 First startup

At first startup, user has to feed in AC power and switch on the manual switch and DC Power Switch. Inverter will start up and LCD display panel will turn on when powered on through AC. Please set language and the correct country (Grid Code) according to your region.
Please make sure that there is no any error, fault or warning showing on home page. Now you can feed in DC power and wait for inverter initially self-test about 2 minutes. If there is enough power generated from PV array, inverter will start feeding in power to grid.

The following ***Figure 4-2*** illustrates the display flow charts of the inverter startup.

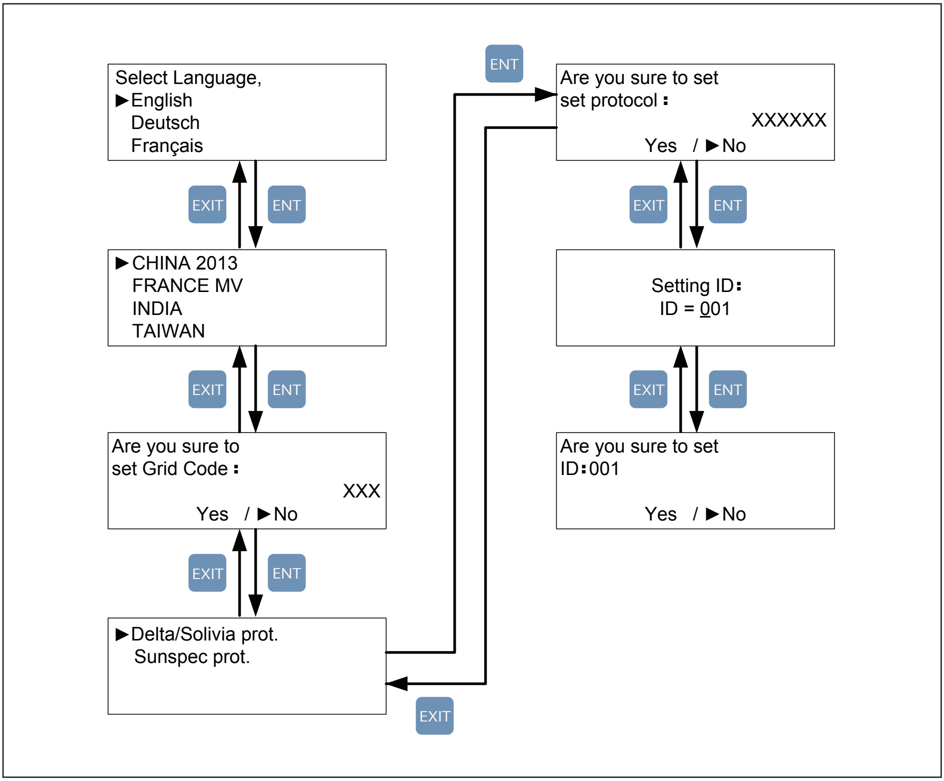


Figure 4-2 : Country, language and ID settings for first startup

4.2.1 Home Page

When inverter is being operated normally, the LCD will display the homepage as shown in ***Figure 4-3***, user can get the information about output power, inverter status, E-today, date and time.

Press "any" key in home page will be directed to the main menu. Press EXIT at main menu or wait 5 minutes without any operation, the display will return to homepage.

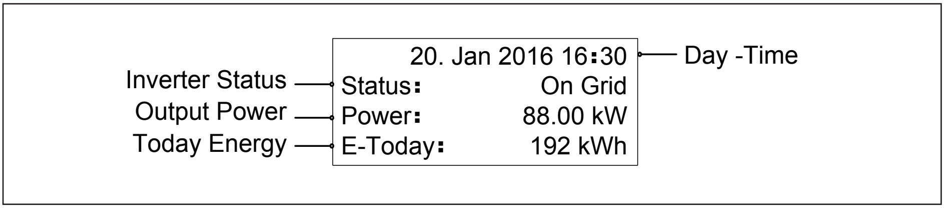


Figure 4-3 : Home page

4.2.2 Power meter / String monitoring

This page displays voltage, current and power from both AC and DC side.

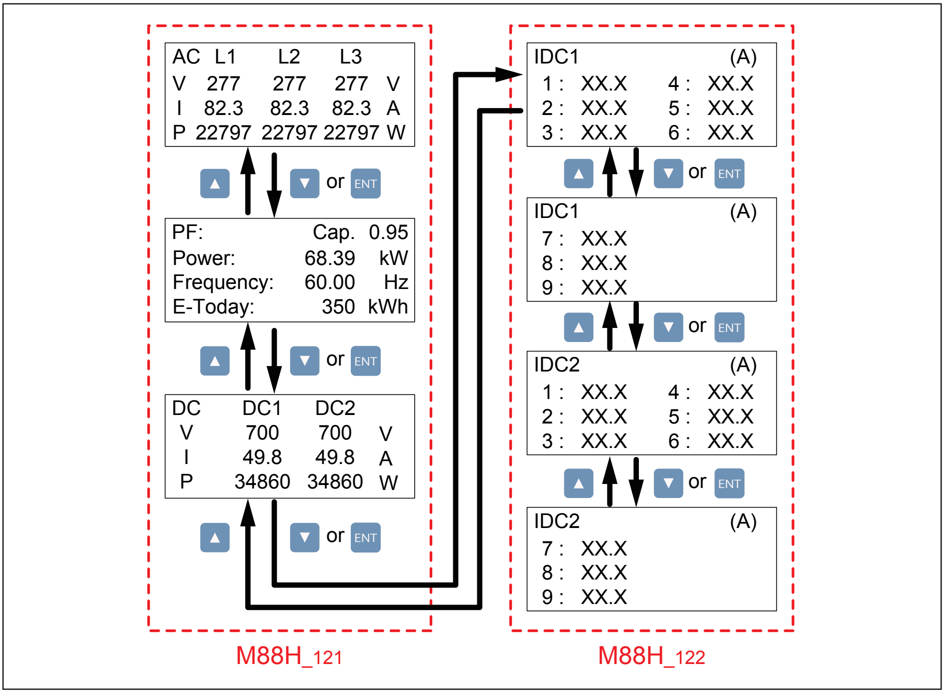


Figure 4-4 : Power meter page

4.2.3 Energy Log

User can view the inverter's life energy and life runtime via Energy Log page.

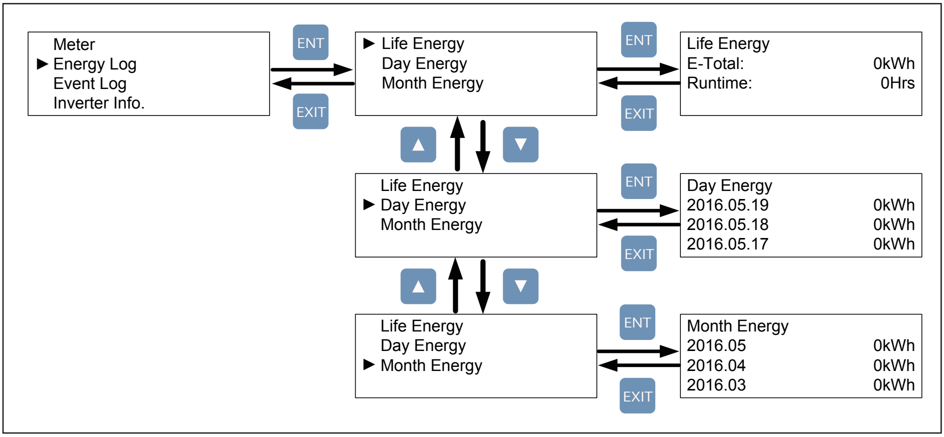


Figure 4-5 : Energy log Page

4.2.4 Event Log

Event Log has two subpages : Error Events page and Grid Report page. Error Events page displays all the events (Error and Fault) and it can show 30 records at a time. Grid Report page only displays the error that occurred at grid side, and it can show 5 records at a time.

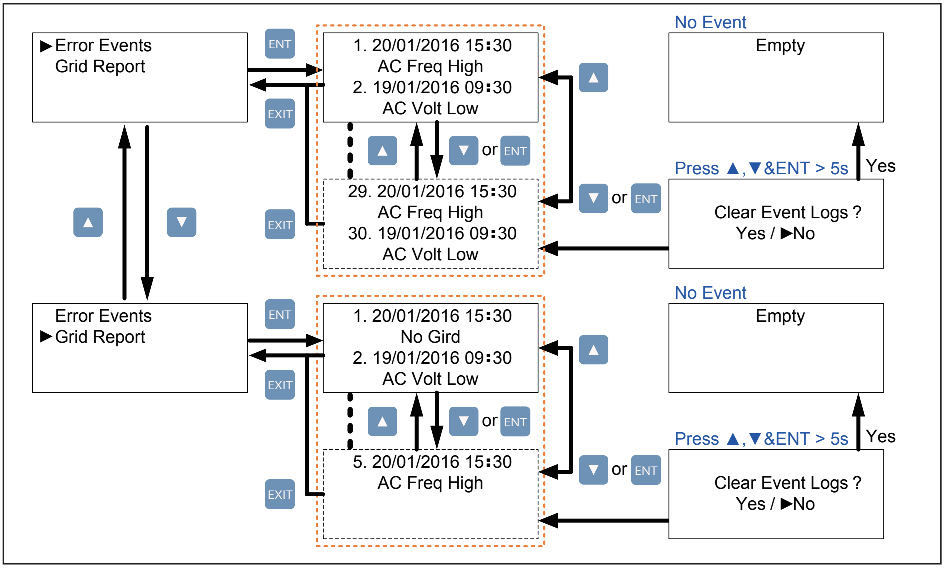


Figure 4-6 : Event log Page

4.2.5 Inverter Information

This page can helps user to recognize the inverter. First section displays serial number, installation date, ID, and firmware version. The settings of inverter functions are described in the following sections.

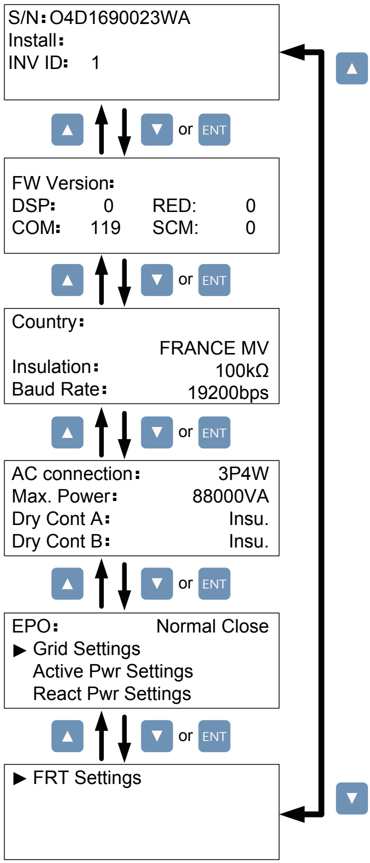


Figure 4-7 : Inverter information page

4.2.6 General Settings

Users can set Language, Date and Time, RS-485 communication baud rate, Protocol and Fan Test in this page.

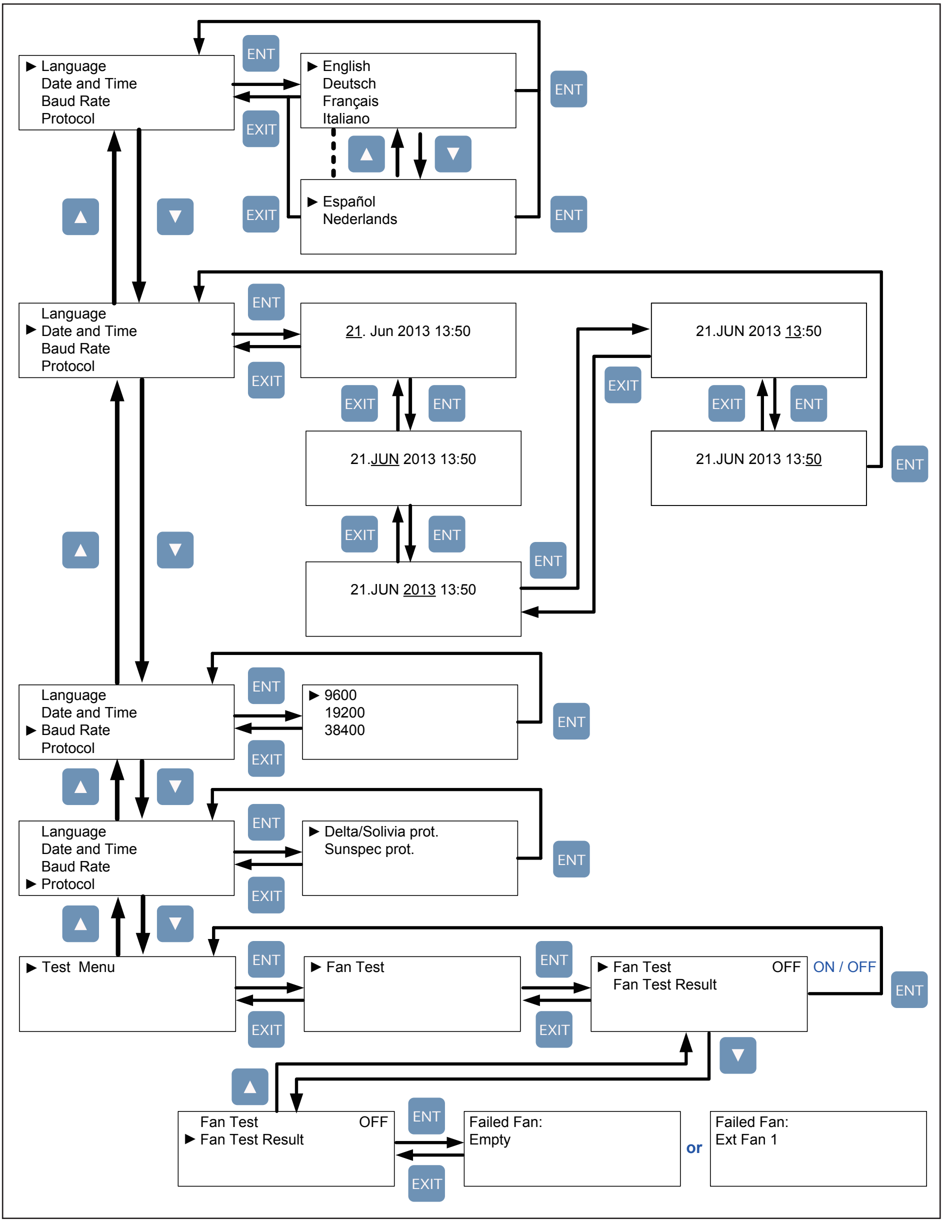


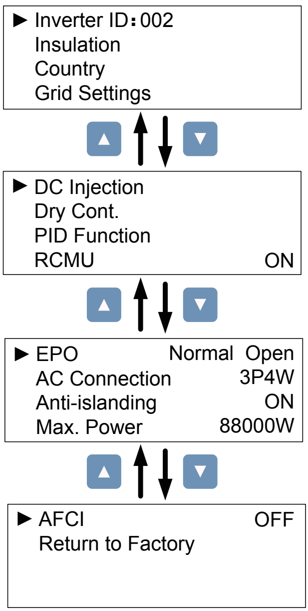
Figure 4-8 : General settings page

4.2.7 Install Settings

CAUTION !

- The settings in Install Settings page can only be adjusted by qualified installers or engineers. Changing these settings may result in damage to the inverter and other equipment.

To enter Install Settings page, users have to enter correct password.
There are 3 sets of password with different permissions: user level, installer level, and manufacturer level. The following sub-sections will introduce the setting items in Install Settings page of user level and installer level.



User Level:
- Inverter ID
- Insulation
- Country
- Dry Cont.
- PID Function
- EPO
- AC Connection
- Max. Power

Installer Level:
- Inverter ID
- Insulation
- Country
- Grid Settings
- Dry Cont.
- PID Function
- EPO
- AC Connection
- Max. Power

Manufacturer Level:
- All Settings

Figure 4-9 : Install settings page

4.2.8 Inverter ID

Inverter ID is used in RS-485 communication, for PC recognizing the inverter. If users connect several inverters together via RS-485, each inverter must have different ID.

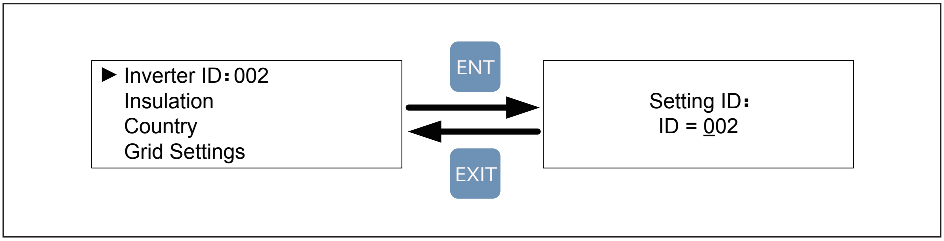


Figure 4-10 : Inverter ID page

4.2.9 Insulation

Before connecting to grid, inverter will measure the impedance between the PV array and PE first. M88H models provide 2 types of impedance measurement methods (ON and OFF) and 2 impedance limits.
Installer must select the appropriate method based on PV array’s wiring.

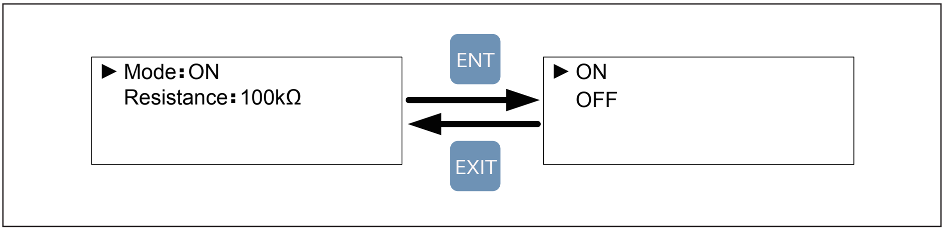


Figure 4-11 : Insulation page

4.2.10 Dry Contact

Users can choose the trigger condition of dry contact. There are 8 options in the setting page: Disable, On Grid, Fan Fail, Insulation, Alarm, Error, Fault, and Warning. Please refer to ***Table 4-2*** for more details about these options.

Setting	Dry Contact Trigger Timing
Disable	No action.
On Grid	Inverter is connecting to grid.
Fan Fail	Fan Fail occurs.
Insulation	Insulation test fail.
Alarm	Any error, fault, or warning occurs.
Error	Any Error occurs.
Fault	Any Fault occurs.
Warning	Any Warning occurs.

Table 4-2 : Dry Contact Trigger Setting

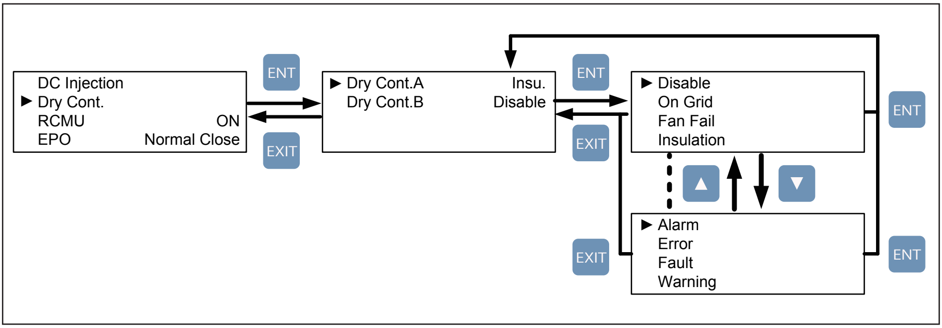


Figure 4-12 : Dry Contact page

4.2.11 PID

The default action time is set with 0, user can set the time from 0-10 or Auto. When PID function is enabled it will be started at 30 minutes after No DC.

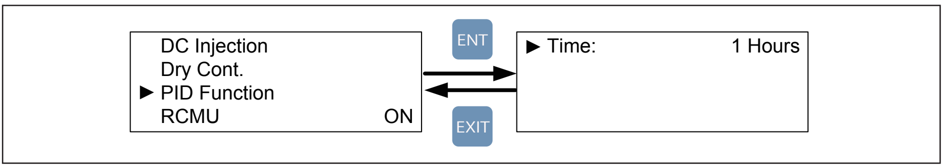


Figure 4-13 : PID function settings

5 Maintenance

Please check the unit regularly. If there are any impaired or loose parts, please contact your solar installer. Ensure that there are no fallen objects in the path of the heat outlet.

WARNING !

- Before any maintenance, please switch AC and DC power off to avoid risk of electrical shock.

5.1 Replace Surge Protection Device (SPD)

M88 series models have the surge protection device (SPD) at both AC and DC side as shown in ***Figure 5-1. Table 5-1*** summarizes the specifications of AC and DC SPD.

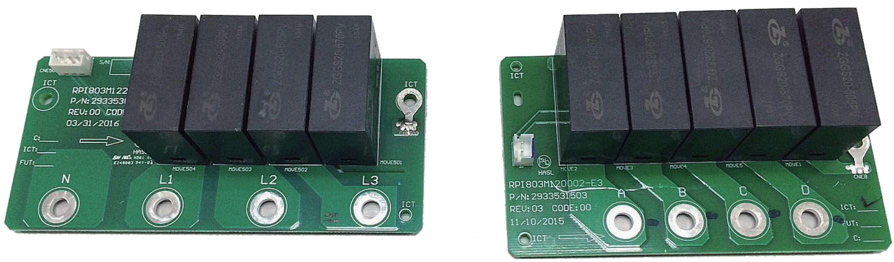


Figure 5-1 : AC and DC SPD

Specification of SPD
Work voltage : 895V (AC RMS value) 1175V (DC) Work Amp In (8/20μs) : 10kA Rate Amp I max (8/20μs) : 20kA Temperature : -40℃～85℃ Manufacturers : Sichuan Zhongguang Lightning Protection Technologies Co., Ltd

Table 5-1 : SPD Specifications

The surge protection devices (SPD), located on both AC and DC input terminals, are designed to protect sensitive circuit elements of the inverter from damage caused by lightning and electrical transient surges. If you find a warning message “AC Surge” or “DC Surge” shown on display panel, please follow the steps below to replace the SPD.

1. Switch AC and DC power off and wait until LCD display turn off.
2. Loosen the 4 screws on the front cover of wiring box compartment.
   You will see AC and DC SPD as indicated in the figure. (***Figure 5-2***)
3. Find out which SPD unit was damaged.
   For AC SPD, “AC Surge” with show on the corner of the LCD panel. (***Figure 5-3***)
   For DC SPD, “DC Surge” with show on the corner of the LCD panel.
4. Pull out the connector (white; 3 pins at AC side; 2 pins at DC side) and replace a whole new SPD PCB. (***Figure 5-4***)
5. Reassemble the inverter. Please be careful of the waterproof seal.

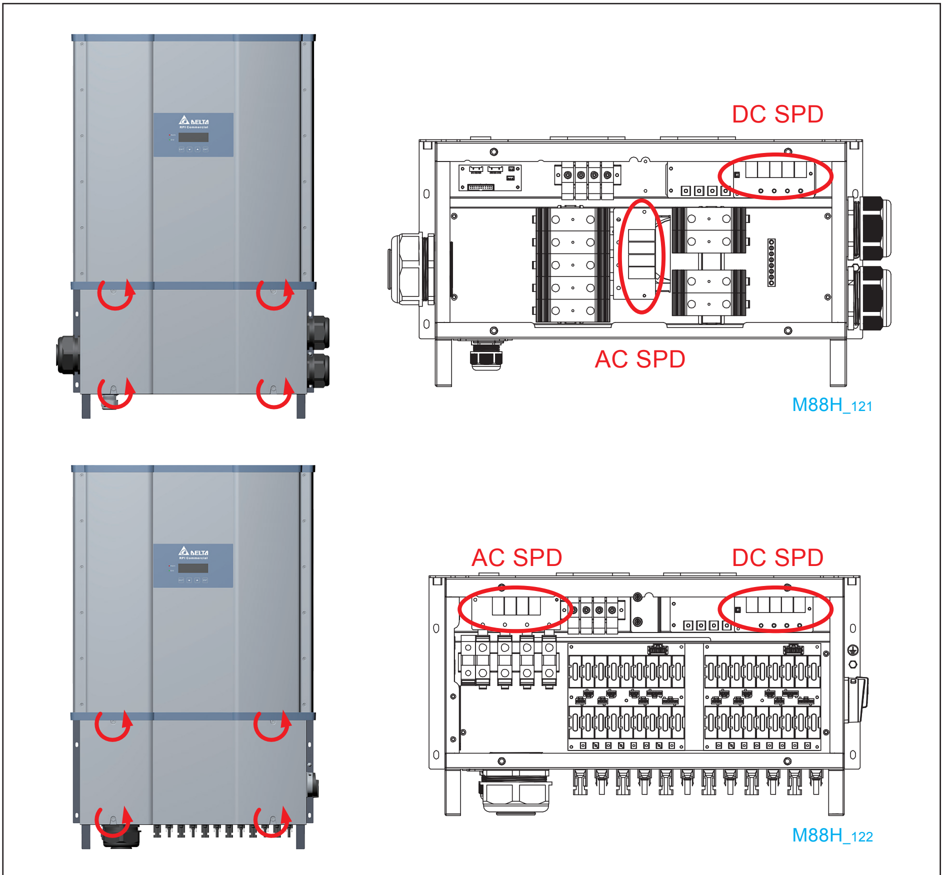


Figure 5-2 : Remove front cover of wiring box compartment

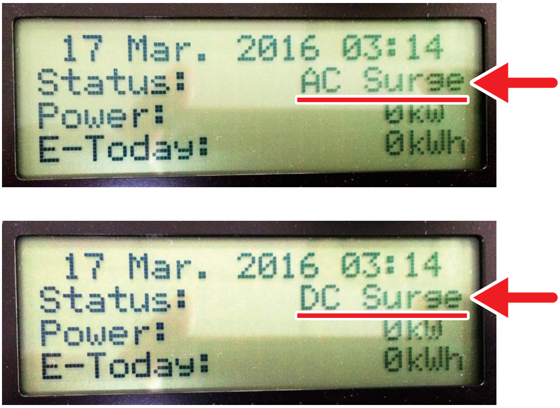


Figure 5-3 : The alarms indicate AC and DC SPD Fail

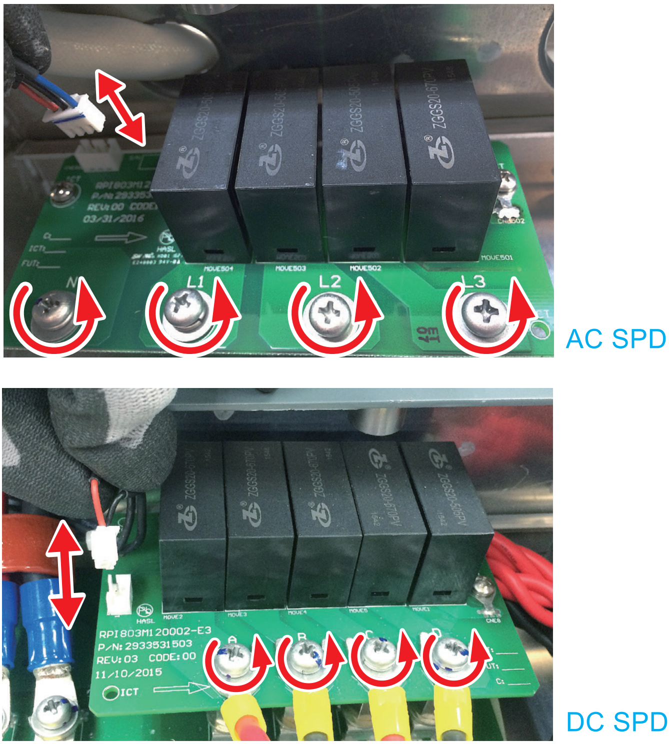


Figure 5-4 : Pull out the connectors and remove the screws as shown in arrows

5.2 Smart Fans Replacement and Filters Cleaning

This section provides the instructions for smart fans replacement and filters cleaning for M88 series models. ***Figure 5-5, 5-6, 5-7*** illustrates the locations of smart fans.

M88 series models have smart fans that can be categorized into two parts: Wiring Box compartment (WB) and Power Module compartment (PM) as shown in the overview picture (***Figure 5-5***). The following procedure dipicts the steps to clean the filters.

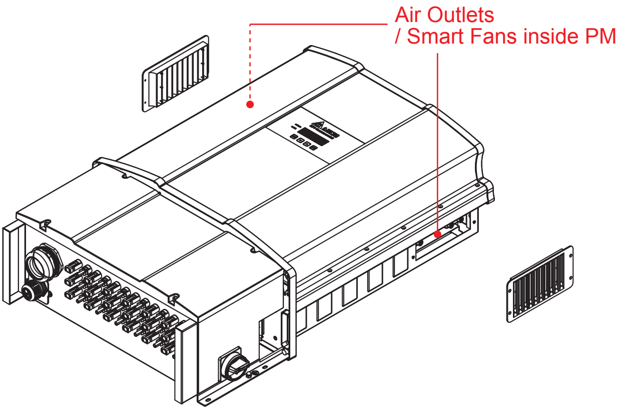


Figure 5-5 : Smart fans location on Power Module compartment

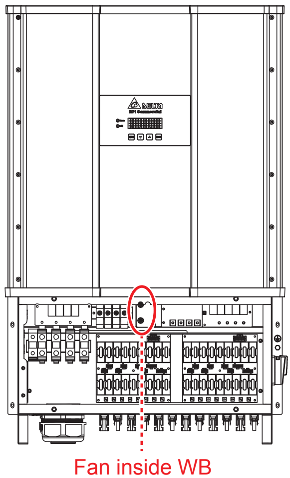


Figure 5-6 : Smart fan location on WB (M88H_122 only)

The M88 series feature 12" potting fans with filter inside the power module compartment. The potting fans are designed long service life that provides high reliability. They feature detection of "FAN-FAIL" alarm as well as power de-rating behavior for safety operation. The Cooling Fan kit is easy to be removed and cleaned. As a result, the replacement of fans is also smart.

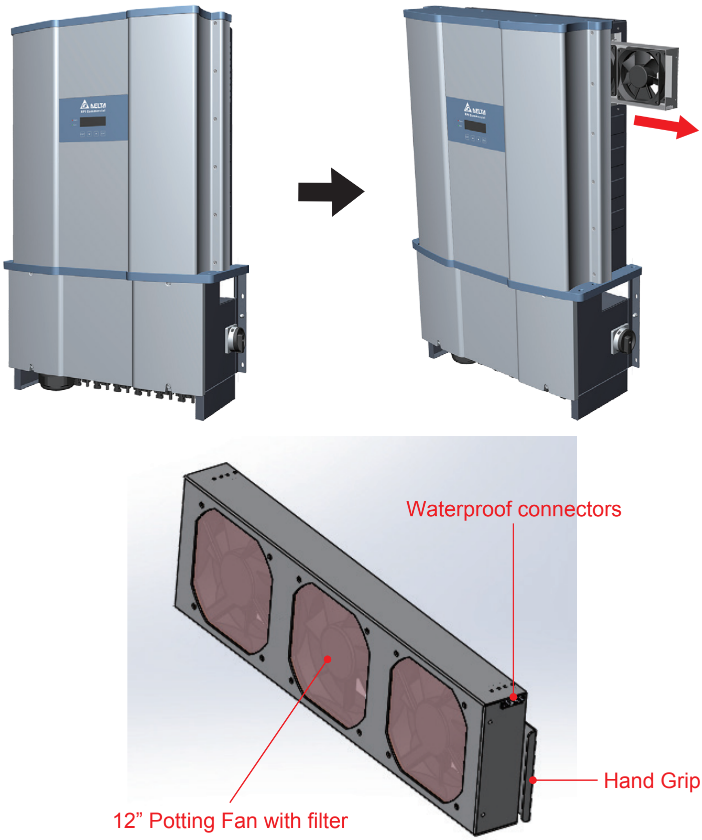


Figure 5-7 : 12" Potting fan kit

ATTENTION

- The smart fans and filter cleaning depends on the quality of the environment.
- The smart fans and filter need to be cleaned every 6 months for their normal use, however,
- The smart fans and filter cleaning is required every 1 month or every quarter of the year, if necessary.

1.Wiring Box compartment (WB): User should disassemble the top 2 pcs screws (2 Black ones as shown in ***Figure 5-8***) outside the fan cabinet and disconnect the connector (white one as shown in ***Figure 5-8***) right in front of the fan cover. Then replace new fan and reassemble the 2 pcs screws and plug in the connector.

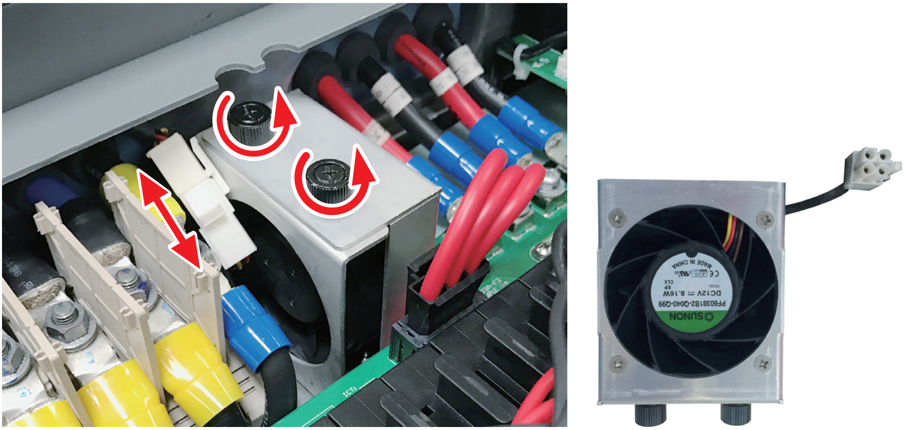


Figure 5-8 : Disassembling fan inside the wiring box compartment (M88H_122 only)

2. Power Module compartment (PM):
Figure 5-9 shows one side (the right-hand side) of the PM compartment only due to the symmetry of the air oulets. Therefore, the following ***Figure 5-9*** shows one side only.

(1) User should disassemble the 4 pcs screws on the air oulet filter panel.
(2) Disconnect the connector (white as shown at the top left corner of second sub-picture) and unscrew the 4 pcs on this side and 4 pcs on the other side (total 8 screws) at the same time.
(3) Make sure both the left 2 and right 2 screws are removed.
(4) Pull out the drawer of the fans.
(5) This sub-picture shows the overview of the smart fans.

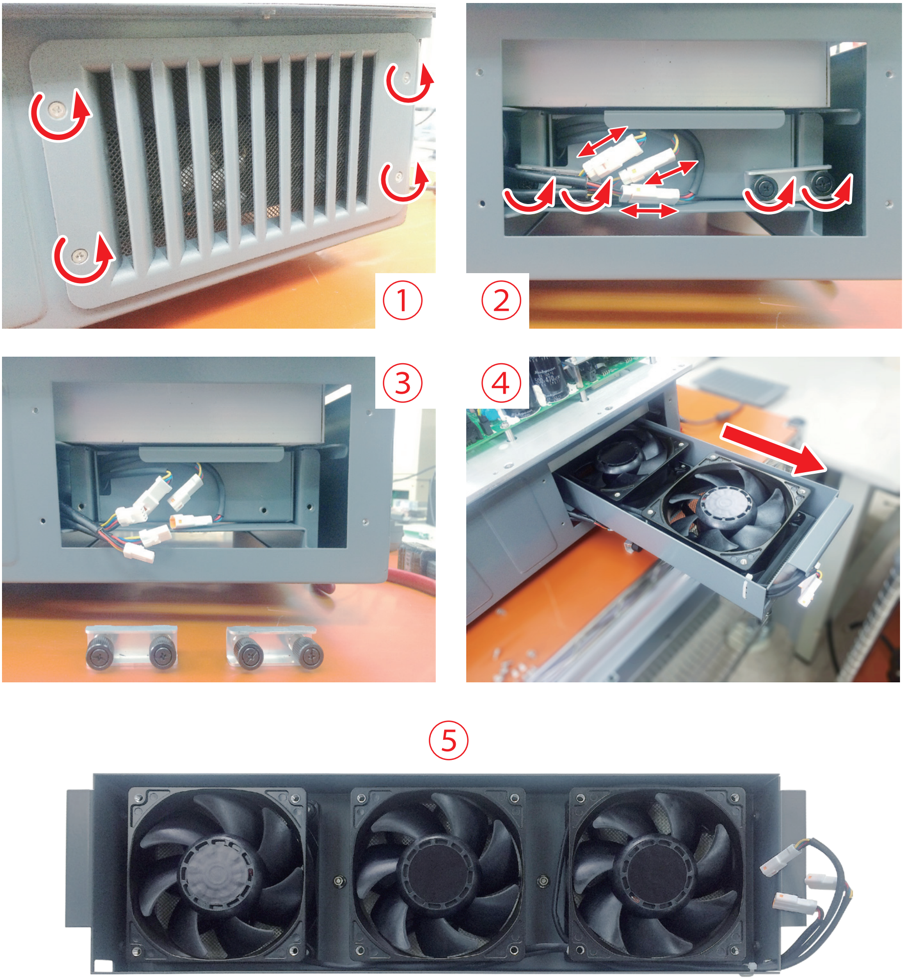


Figure 5-9 : Disassembling fans inside the power module compartment (showing one side only)

Figure 5-10 illustrates filters location of M88 series models.

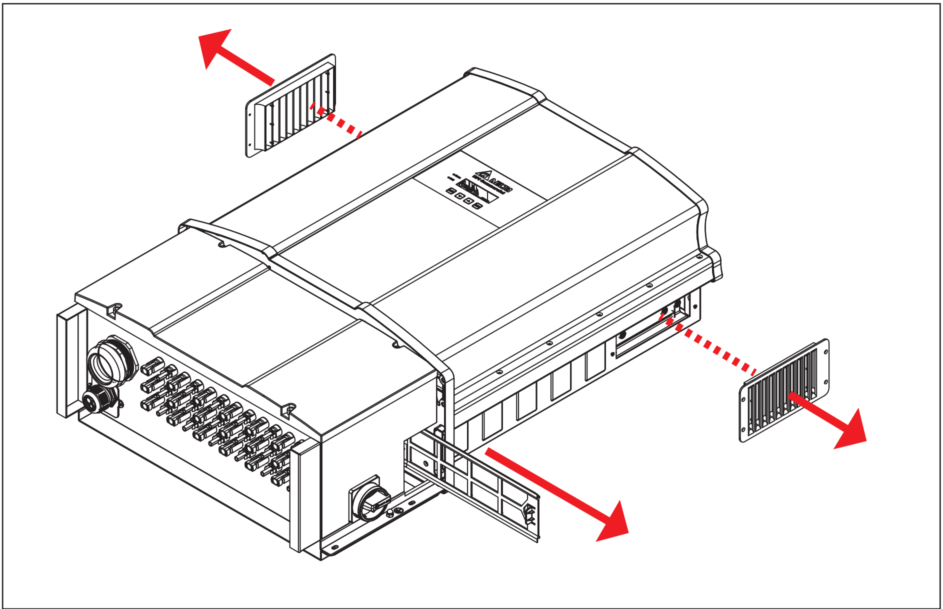


Figure 5-10 : Removal of filters

5.3 De-Commissioning

If it is necessary to put the device out of operation for maintenance and / or storage, please follow the instructions below.

DANGER : ELECTRICAL HARZARD!!

To avoid any serious injuries, please follow the procedures:
- Switch off Manual Switch to shut down the inverter.
- Switch off AC circuit breaker to disconnect with electricity grid.
- Switch off the PV array switch to disconnect from the PV array.
- Use proper voltmeter to confirm that the AC and DC power are disconnected from the unit.
- Remove the AC terminal immediately to completely disconnect from electricity grid.
- Remove the DC terminals to disconnect from PV Array.
- Remove the communication module RS-485 connection from the computer.

CAUTION : HOT SURFACES, DO NOT TOUCH !

- Please be careful of hot surfaces if the inverter is just shutting down.
- Do not perform any task until the product cool down sufficiently.

CAUTION : POSSIBILE INJURY !

- The inverter weighs more than 80 kg (177 lb). The risk of injury may happen when the inverter is carried incorrectly or dropped during transported or when attaching or removing it from the wall mounting bracket.

ATTENTION

- Please be careful of the screws and nuts after removing them. Do not leave them at any corner inside the wiring box compartment.

5.3.1 Disassemble the Wiring box compartment

In order to decommission the inverter of its service life, please follow the following instructions.

In order to disassemble the wiring box compartment (WB) if neccessary, please follow the following instructions.

1. Please make sure turning off the external AC breaker and DC switches
2. Double-check the inverter is shutted down and there is no electrical hazard.
3. ***Figure 5-11*** shows the correct way to turn off the DC swiches.

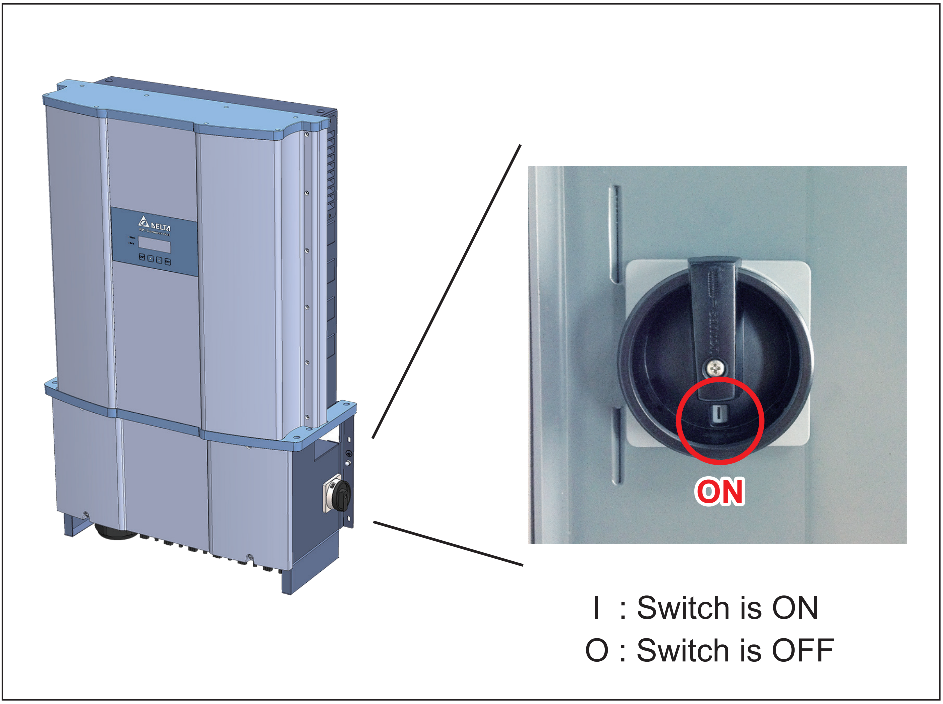


Figure 5-11 : The ON/OFF positions of DC Switches (M88H_122 only)

Please confirm both the AC and DC power is turned off.

4. Open the wiring box compartment lid.
5. Then remove AC, DC, communication cables.
6. Unscrew those screws shown in ***Figure 5-12.***

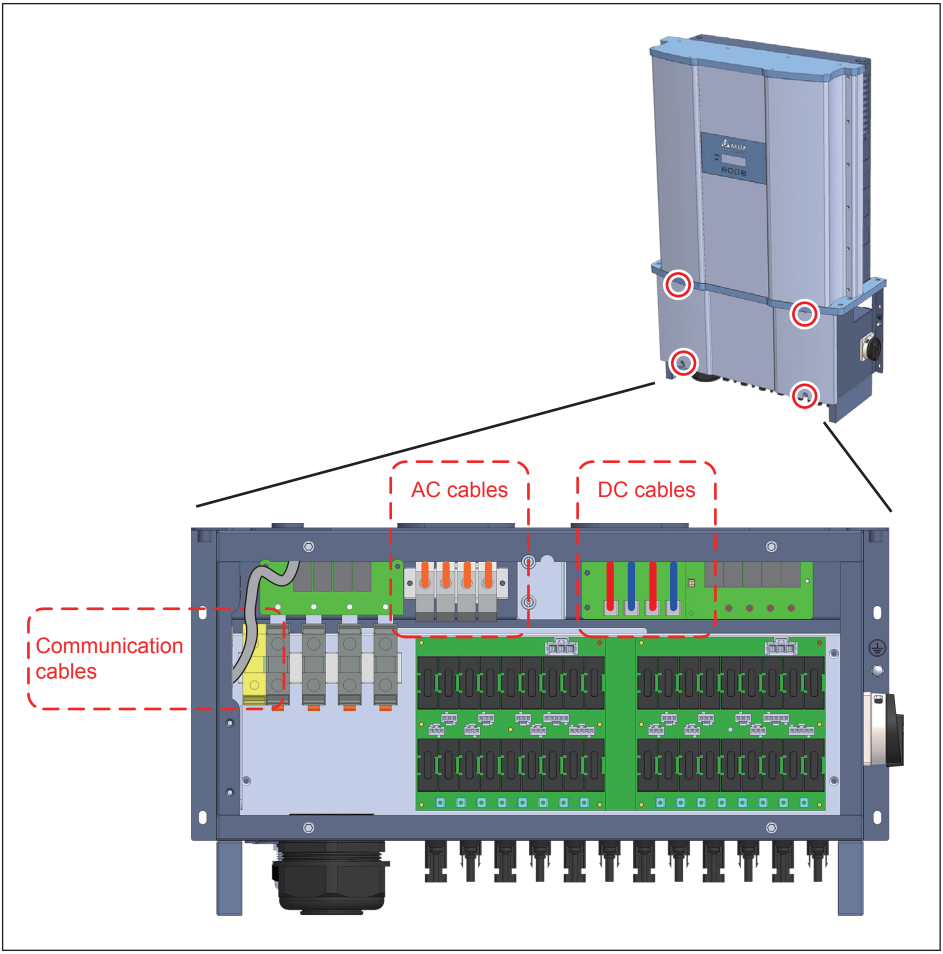


Figure 5-12 : Remove AC, DC, and communication cables and then unscrew 6 screws

7. Unscrew 8 screws and use hands to grip tightly to separate the wiring box compartment from the power module compartment as shown in ***Figure 5-13.***
8. Make sure the packing has been installed.

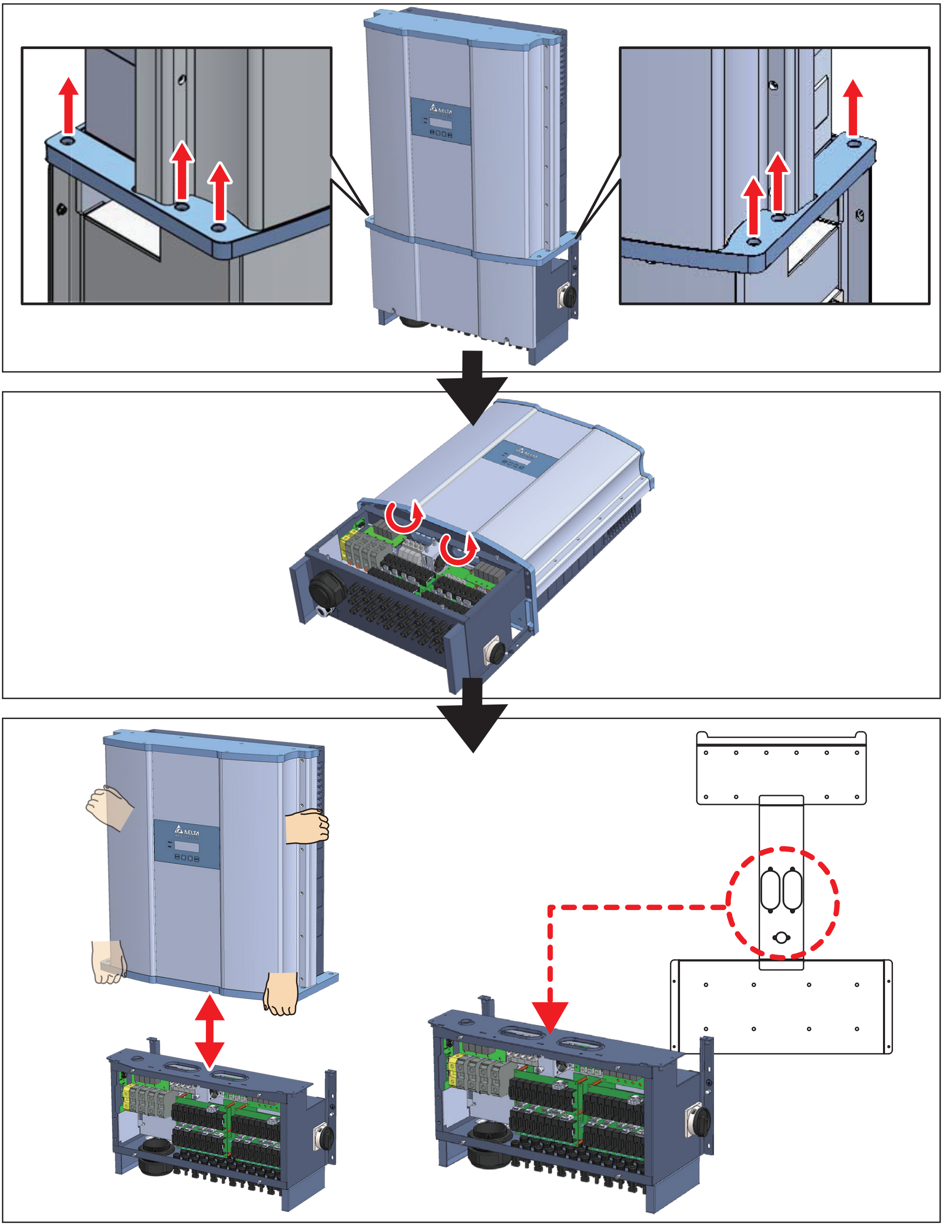


Figure 5-13 : Remove 8 screws and then separate WB from PM

6 Error message and Trouble Shooting

ERROR		
Message	**Possible cause**	**Action**
AC Freq High (E01)	1. Actual utility frequency is over the OFR setting 2. Incorrect country setting 3. Detection circuit malfunction	1. Check the utility frequency 2. Check country setting 3. Contact our customer service for technical support
AC Freq Low (E02)	1. Actual utility frequency is under the UFR setting 2. Incorrect country or Grid setting 3. Detection circuit malfunction	1. Check the utility frequency 2. Check country & Grid setting 3. Contact our customer service for technical support
Grid Quality (E07)	Non-linear load in Grid and near to inverter	Grid connection of inverter need to be far away from non-linear load if necessary
HW Con. Fail (E08)	1. Wrong connection in AC terminal 2. Detection circuit malfunction	1. Check the AC connection in accordance with the user manual 2. Contact our customer service for technical support
No Grid (E09)	1. AC breaker is OFF 2. Disconnect in AC terminal	1. Switch on AC breaker 2. Check the connection in AC terminal and make sure it connects to inverter
AC Volt Low (E10, E15, E20)	1. Actual utility voltage is under the UVR setting 2. Incorrect country or Grid setting 3. Wrong connections in AC terminal 4. Detection circuit malfunction	1. Check the utility voltage within the suitable range 2. Check country & Grid setting 3. Check the connection in AC terminal
AC Volt High (E11, E13, E16, E18, E21, E23)	1. Actual utility voltage is over the OVR setting 2. Utility voltage is over the Slow OVR setting during operation 3. Incorrect country or Grid setting 4. Detection circuit malfunction	1. Check the utility voltage within the suitable range 2. Check country & Grid setting 3. Contact our customer service for technical support
Solar1 High (E30)	1. Actual Solar1 voltage is over 1000Vdc 2. Detection circuit malfunction	1. Modify the solar array setting, and make the Voc less than 1000Vdc 2. Contact our customer service for technical support

ERROR		
Message	**Possible cause**	**Action**
Solar2 High (E31)	1. Actual Solar2 voltage is over 1000Vdc 2. Detection circuit malfunction	1. Modify the solar array setting, and make the Voc less than 1000Vdc 2. Contact our customer service for technical support
Insulation (E34)	1. PV array insulation fault 2. Large PV array capacitance between Plus to Ground or Minus to Ground or both. 3. Detection circuit malfunction	1. Check the insulation of Solar inputs 2. Check the capacitance, dry PV panel if necessary 3. Contact our customer service for technical support

Table 6-1 : Error Message

Warning		
Message	**Possible cause**	**Action**
Solar1 Low (W01)	1. Actual Solar1 voltage is under the limit 2. Some devices were damaged inside the inverter if the actual Solar1 voltage is close to "0" 3. Detection circuit malfunction	1. Check the Solar1 voltage connection to the inverter terminal 2. Check all switching devices in boost1 3. Contact our customer service for technical support
Solar2 Low (W02)	1. Actual Solar2 voltage is under the limit 2. Some devices were damaged inside the inverter if the actual Solar2 voltage is close to "0" 3. Detection circuit malfunction	1. Check the Solar2 voltage connection to the inverter terminal 2. Check all switching devices in boost2 3. Contact our customer service for technical support
HW FAN (W11)	1. One or more fans are locked 2. One or more fans are defective 3. One ore more fans are disconnected 4. Detection circuit malfunction	1. Remove the object that stuck in the fan(s) 2. Replace the defective fan(s) 3. Check the connections of all fans 4. Contact our customer service for technical support
AC Surge DC Surge	1. Inverter was struck by lighting. 2. One or more SPD are defective 3. One or more SPD are disconnected 4. Detection circuit malfunction	1. Check inverter’s status 2. Replace the defective SPD 3. Check the connections of SPDs 4. Contact our customer service for technical support

Table 6-2 : Warning Message

FAULT		
Message	**Possible cause**	**Action**
DC Injection (F01, F02, F03)	1. Utility waveform is abnormal 2. Detection circuit malfunction	1. Check the utility waveform. Grid connection of inverter need to be far away from non-linear load if necessary 2. Contact our customer service for technical support
Temperature (F05)	1. The ambient is over 60℃ (The installation is abnormal) 2. Detection circuit malfunction	1. Check the installation ambient and environment 2. Contact our customer service for technical support
Temperature (F07)	1. Ambient temperature is <-30 ℃ 2. Detection circuit malfunction	1. Check the installation ambient and environment 2. Contact our customer service for technical support
HW NTC1 Fail (F06)	1. Ambient temperature >90 ℃ or <-30 ℃ 2. Detection circuit malfunction	1. Check the installation ambient and environment 2. Contact our customer service for technical support
HW NTC2 Fail (F08)	1. Ambient temperature >90 ℃ or <-30 ℃ 2. Detection circuit malfunction	1. Check the installation ambient and environment 2. Please contact our customer service
HW NTC3 Fail (F09)	1. Ambient temperature >90 ℃ or <-30 ℃ 2. Detection circuit malfunction	1. Check the installation ambient and environment 2. Contact our customer service for technical support
HW NTC4 Fail (F10)	1. Ambient temperature >90 ℃ or <-30 ℃ 2. Detection circuit malfunction	1. Check the installation ambient and environment 2. Contact our customer service for technical support
Relay Test Open (F13)	1. Driver circuit for relay is defective 2. Relay(s) is defective 3. Detection circuit malfunction (Inverter voltage)	1. Check the input voltage, must >150Vdc 2. Replace the defective relay 3. Contact our customer service for technical support
HW DSP ADC1 (F15)	1. Insufficient input power 2. Auxiliary power circuitry malfunction 3. Detection circuit malfunction	1. Check the input voltage, must >150Vdc 2. Check the auxiliary circuitry inside the inverter 3. Contact our customer service for technical support

FAULT		
Message	**Possible cause**	**Action**
HW DSP ADC2 (F16)	1. Insufficient input power 2. Auxiliary power circuitry malfunction 3. Detection circuit malfunction	1. Check the input voltage, must >150Vdc 2. Check the auxiliary circuitry inside the inverter 3. Contact our customer service for technical support
HW DSP ADC3 (F17)	1. Insufficient input power 2. Auxiliary power circuitry malfunction 3. Detection circuit malfunction	1. Check the input voltage, must >150Vdc 2. Check the auxiliary circuitry inside the inverter 3. Contact our customer service for technical support
HW Red ADC1 (F18)	1. Insufficient input power 2. Auxiliary power circuitry malfunction 3. Detection circuit malfunction	1. Check the input voltage, must >150Vdc 2. Check the auxiliary circuitry inside the inverter 3. Contact our customer service for technical support
HW Red ADC2 (F19)	1. Insufficient input power 2. Auxiliary power circuitry malfunction 3. Detection circuit malfunction	1. Check the input voltage, must >150Vdc 2. Check the auxiliary circuitry inside the inverter 3. Contact our customer service for technical support
HW Eff. (F20)	1. The calibration is incorrect 2. Current feedback circuit is defective	1. Check the accuracy of current and power 2. Check the current feedback circuit inside the inverter
HW COMM1 (F23)	1. DSP is idling 2. The communication connection is disconnected 3. The communication circuit is malfunction	1. Contact our customer service for technical support 2. Check the connection interface RS-485 3. Check the communication card
HW COMM2 (F22)	1. Red. CPU is idling 2. The internal communication connection is disconnected	Contact our customer service for technical support
Ground Cur. (F24)	1. PV array insulation fault 2. Large PV array capacitance between Plus to Ground or Minus to Ground 3. Either side of boost driver or boost choke malfunction 4. Detection circuit malfunction	1. Check the insulation of Solar inputs 2. Check the capacitance (+ <-> GND & - <-> GND), must < 2.5uF. Install a external transformer if necessary 3. Contact our customer service for technical support

FAULT		
Message	**Possible cause**	**Action**
HW Con. Fail (F26)	1. Power line is disconnected inside the inverter 2. Current feedback circuit is defective	1. Check the power lines inside the inverter 2. Contact our customer service for technical support
RCMU Fail (F27)	1. RCMU is disconnected 2. Detection circuit malfunction	1. Check the RCMU connection inside the inverter 2. Contact our customer service for technical support
RLY Short (F28)	1. One or more relays are sticking 2. The driver circuit for the relay malfunction	Contact our customer service for technical support
RLY Open (F13, F29)	1. One or more relays are abnormal 2. The driver circuit for the relay malfunction 3. The detection accuracy is not correct for Vgrid and Vout	Contact our customer service for technical support
Bus Unbal. (F30)	1. Not totally independent or parallel between inputs 2. PV Array short to Ground 3. Driver for boost is defective or disconnected 4. Detection circuit malfunction	1. Check the inputs connections 2. Check the PV Array insulation 3. Contact our customer service for technical support
HW Bus OVR (F31, F33, F35)	1. Driver for boost is defective 2. Voc of PV array is over 1000Vdc 3. Surge occurs during operation 4. Detection circuit malfunction	Contact our customer service for technical support
AC Cur. High (F36, F37, F38, F39, F40, F41)	1. Surge occurs during operation 2. Driver for inverter stage is defective 3. Switching device is defective 4. Detection circuit malfunction	Contact our customer service for technical support
HW CT A Fail (F42)	1. Test current loop is broken 2. CTP3 is defective 3. Detection circuit malfunction	Contact our customer service for technical support

FAULT		
Message	**Possible cause**	**Action**
HW CT B Fail (F43)	1. Test current loop is broken 2. CTP4 is defective 3. Detection circuit malfunction	Contact our customer service for technical support
HW CT C Fail (F44)	1. Test current loop is broken 2. CTP5 is defective 3. Detection circuit malfunction	Contact our customer service for technical support
HW AC OCR (F45)	1. Large Grid harmonics 2. Switching device is defective 3. Detection circuit malfunction	1. Check the utility waveform. Grid connection of inverter need to be far away from non-linear load if necessary 2. Check all switching devices in inverter stage 3. Contact our customer service for technical support
DC Cur. High (F60, F61, F70, F71)	1. Switching device in boost is defective 2. Driver for boost is defective 3. Input current detection circuit malfunction	1. Check all switching device in boost 2. Check the driver circuit for boost inside the inverter 3. Check input current detection circuit
HW DC RLY (F76)	One or more DC relays are abnormal	Please contact our customer service for technical support

Table 6-3 : Fault Message

7 Technical Data

Model	M88H_121	M88H_122
DC Input		
Max. input power	Vac230/400V：76kW Vac277/480V：91kW	
Recommended PV power	Vac230/400V：90kW Vac277/480V：110W	
Occasionally maximum voltage	1100 V *	
Operating voltage range	200 - 1000 V	
Start voltage	> 250 V	
MPP voltage, rated power	Vac230/400V：500-800V Vac230/400V：600-800V	
Rated voltage	Vac230/400V：595V Vac277/480V：710V	
MPP tracker	2	
Max. input current / Each MPPT	140 / 70A	
Max. Isc / Each MPPT	180 / 90A	
Connection type	2-250kcmil terminal block for 2 MPPTs	18 pairs of MC4 connector
Type II SPD	●	●
15A string fuses	—	●
DC switch	—	●
AC Output		
Max. output power	Vac230/400V：73kW Vac277/480V：88kW	
Max. output current	106A	
Inrush current	40A / 100μs	
Maximum output fault current (rms)	115.4A (rms)	
Maximum output over current protection	125A	
Rated voltage	3Ph, 230/400 & 277/480Vac	
Operating voltage range	Vac230/400V：±30% Vac277/480V：±20%	
Operating frequency range	50/60Hz ± 5Hz	
Power factor	1 at rated power, 0.8 ind ~ 0.8 cap adjustable	
Protection	Type II SPD	
THD	< 3%	
Connection type	50 ~120 mm^2	35 ~95 mm^2
Night time consumption	< 3 W	

* The max withstand voltage is 1100Vdc. (the inverter stops to operate when the PV voltage is over 1000Vdc)
About 1000Vdc above application, please refer to appendix.

● : Available
— : Not Available

Model	M88H_121	M88H_122
Efficiency		
Peak Effieiency	98.8 %	
Euro Effieiency	98.5 %	
Information		
Communication port	RS-485	
Display	20 x 4 LCD	
Regulation		
	IEC 62109-1/-2 VDE-AR-N 4105 EN 61000-6-1 EN 61000-6-2 EN 61000-6-3 EN 61000-6-4 CE compliance	
General Data		
Operating temp. range	-25~60℃ (Max power: -25~35℃)	
Protection Level	IP65	
Operating elevation	< 3000 m	
Cooling	Forced air cooling plus Smart Fans control	
Dimension (W x H x D) (mm)	615 x 962 x 275	
Weight (kg)	84	
Noise (1m)	75.8 dB	
Overvoltage category	AC output :III, DC Input :II	
Maximum backfeed current to the array	0	
Protective Class	I	
Pollution Degree	3	
Humidity range	4-100%	

Table 7-1 : Specifications for M88H

Smarter. Greener. Together.

Version 1.02 | 2016 | Oct.